

Reviewed Preprint

v1 • June 20, 2025

Not revised

Reviewed Preprint

v2 • February 25, 2026

Revised by authors

For correspondence:

mc115@cam.ac.uk

* These authors made equal contributions

Competing interests: No competing interests declared

Funding: See [page 34](#)

Reviewing editor: Christine Clayton, Centre for Molecular Biology of Heidelberg University (ZMBH), Germany

© 2025, Banerjee et al. This article is distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use and redistribution provided that the original author and source are credited.

Cell surface localisation of GPI-anchored receptors in *Trypanosoma brucei*

Sourav Banerjee^{1,*}, Nicola Minshall^{1,*}, Alexander D Cook^{2,3}, Olivia Macleod¹, Helena Webb¹, Matthew Higgins^{2,3}, Mark Carrington^{1,*} ✉

¹Department of Biochemistry, University of Cambridge, Cambridge, United Kingdom • ²Department of Biochemistry, University of Oxford, Oxford, United Kingdom • ³Kavli Institute for Nanoscience Discovery, Dorothy Crowfoot Hodgkin Building, University of Oxford, Oxford, United Kingdom

eLife Assessment

This **valuable** manuscript investigates the localisation of nutrient receptors in bloodstream stage trypanosomes, with implications for both nutrient uptake and immune evasion. Results after direct fixation of the cells in culture medium (as opposed to fixation after centrifugation) provide **compelling** evidence that the amounts of receptors on the surface of the cell, as opposed to the flagellar pocket, have previously been severely underestimated.

<https://doi.org/10.7554/eLife.107191.2.sa3>

Abstract

Trypanosoma brucei, the causal agent of Human and Animal African trypanosomiasis proliferates in the extracellular milieu of mammals. It acquires host macromolecular nutrients, by receptor mediated endocytosis. The best characterised cell surface receptor is for transferrin (TfR) and it has been reported to be preferentially localised in the flagellar pocket domain of the plasma membrane, the sole site of endocytosis. In this location the TfR may be inaccessible to adaptive immune system effectors. The *T. brucei* genome encodes ~15 TfR variants, and here we compared two, the first attached to the plasma membrane by a single glycosylphosphatidylinositol (GPI)-anchor and the other by two. Transferrin uptake kinetics were similar and rapid for both. Unexpectedly, initial binding of transferrin occurred over the whole cell surface suggesting the TfR was not localised solely in the flagellar pocket. This localisation was confirmed by immunofluorescence assays and was independent of the number of GPI-anchors. Two other GPI-anchored receptors were investigated to determine whether localisation to the whole cell surface was a general property of GPI-anchored receptors. Haptoglobin-haemoglobin uptake assays and immunofluorescence localisation of complement factor H receptor showed both were also whole cell surface localised. The mechanisms by which trypanosome receptors are protected from antibody-mediated attack are more complex than hiding in a pocket.

Introduction

Trypanosoma brucei proliferates within the bloodstream and tissue spaces of mammals and can maintain a long-term infection lasting in some cases for years. To do this, it must avoid the host adaptive and innate immune response at the same time as acquiring nutrients from the host. The adaptive immune response is overcome through a population survival strategy based on a densely packed surface coat of variant surface glycoprotein (VSG) that undergoes antigenic variation. As the antibody titre against one VSG increases, a small subset of trypanosomes that have switched expression to a different VSG escape until they are recognised in turn and so on in an iterative process (Mugnier et al., 2015 [↗](#)). The trypanosome has evolved mechanisms to increase the titre of

host immunoglobulin required for killing. First, there is rapid endocytosis and degradation of VSG bound antibody (Engstler et al., 2007 [\[1\]](#)) and second, a receptor for IgG Fc region (ISG75) (Mikkelsen et al., 2024 [\[2\]](#)). How the innate immune system is countered is less well understood (Pinger et al., 2018 [\[3\]](#)) but receptors for Complement C3, known as ISG65, and Factor H (FHR) have been identified and characterised (Cook et al., 2024 [\[4\]](#); Macleod et al., 2020 [\[5\]](#), 2022 [\[6\]](#)). Acquisition of small molecule nutrients, such as glucose and amino acids, use typical eukaryotic membrane transporters.

Obtaining host macromolecule nutrients poses a more complex problem. First, *T. brucei* has a very wide host range and presumably receptors recognise ligands from a range of host species, this has been shown to be the case for the transferrin (TfR) (Trevor et al., 2019 [\[7\]](#)).

Second, as the TfR and haptoglobin haemoglobin receptor (HpHbR) (Lane-Serff et al., 2016 [\[8\]](#)) bind ligands that have similar dimensions to immunoglobulin molecules then why are host antibodies against receptors not sufficient to clear an infection? Some models to account for this suggested that the VSG coat acts as a barrier preventing immunoglobulin access.

However, it was shown that antibodies could penetrate a significant distance into the VSG coat (Hsia et al., 1996 [\[9\]](#)) but not to the small plasma membrane proximal VSG domain (Schwede et al., 2011 [\[10\]](#)). Both the ligand size and the penetration of antibodies into the VSG coat made it unlikely that receptors were sterically protected from antibody binding.

Immunization of mice with recombinant ISG65 or ISG75 did not protect against infection (Ziegelbauer & Overath, 1993 [\[11\]](#)). In addition, accessibility of the HpHbR was demonstrated by the ability of a monoclonal antibody drug conjugate to clear infections in a mouse model (MacGregor et al., 2019).

The trypanosome flagellar pocket is an invagination of the plasma membrane at the base of the single flagellum and is the sole site of endo- and exocytosis. The location of the flagellar pocket at the posterior end of the cell is central to the mechanism of clearance of immunoglobulin molecules bound to VSG. The forward swimming motion of the trypanosome exerts a hydrodynamic force on the immunoglobulin that protrudes above the top of the VSG coat pushing the antibody-VSG complex towards the flagellar pocket and subsequent endocytosis (Engstler et al., 2007 [\[1\]](#)). Since the hydrodynamic flow requires only binding to a cell surface protein, it is likely to occur with any large ligand bound to a receptor providing it has free lateral movement in the plane of the membrane. The flagellar pocket is a discrete domain of the plasma membrane (Gadelha et al., 2009 [\[12\]](#)) with several exclusively localised proteins (Gadelha et al., 2015 [\[13\]](#)). At the neck of the pocket, there is a restriction introduced by close contact between the plasma membranes of the flagellum and pocket as it emerges onto the cell body. This flagellar pocket collar is maintained by specific proteins (Bonhivers et al., 2008 [\[14\]](#)) (Gadelha et al., 2009 [\[12\]](#)). The exclusive localisation of some proteins to the plasma membrane of the flagellar pocket, others to the cell body, and a range to both, indicates that there is regulated passage between the two domains of the plasma membrane.

There is variation in the reported localisation of receptors to different domains of the plasma membrane. TfR was the first receptor to be identified, it is glycosylphosphatidylinositol (GPI)-anchored and has been localised to the flagellar pocket and endosomal membrane in some studies but others suggested it was on the flagellar pocket, cell body plasma membrane and endosomal membrane (Duncan et al., 2024 [\[15\]](#); Mussmann et al., 2003 [\[16\]](#), 2004 [\[17\]](#); Steverding et al., 1994a [\[18\]](#), 1995 [\[19\]](#); Tiengwe et al., 2017 [\[20\]](#)). The Complement Factor H receptor (FHR) is a monomer probably GPI-anchored and is localised to the cell body and flagellar plasma membrane (Macleod et al., 2020 [\[5\]](#)). HpHbR, also probably GPI-anchored, has been localised to the flagellar pocket (Vanhollebeke et al., 2008 [\[21\]](#)). Reporter proteins with a single GPI-anchor are also present on the cell body plasma membrane (Schwartz et al. 2005 [\[22\]](#)). ISG65 and ISG75 are type I transmembrane proteins and localise to the flagellar, cell body, and pocket plasma membrane as well as the endosome (Leung et al., 2011 [\[23\]](#); Ziegelbauer et al., 1992 [\[24\]](#)). The localisation of receptors to the flagellar pocket has been proposed to protect the cell from downstream effects of antibody

recognition of the receptor but the discovery of new receptors, such as Complement C3 binding by ISG65 (Macleod et al., 2022 [DOI](#)) and FHR (Macleod et al., 2020 [DOI](#)) that are more abundant and distributed over the cell surface seems to contradict this argument.

Newly synthesised or recycled proteins are added to the plasma membrane at the flagellar pocket. Some proteins are retained in the flagellar pocket and have limited or no distribution over the cell surface (Billington et al., 2023 [DOI](#)). It is assumed that these proteins have a retention and/or retrieval signal. For transmembrane proteins a signal could be in the cytoplasm. In contrast, GPI-anchored proteins have no cytoplasmic component and recognition must occur on the external face of the plasma membrane. Any signal for retention has to operate in a background of sequence variation in VSGs which makes short amino acid motifs unlikely. TfR is a heterodimer with the two subunits, ESAG6 and ESAG7, forming a single binding site for monomeric transferrin. ESAG6, but not ESAG7, has a GPI-anchor so the TfR is tethered to the plasma membrane by a single anchor in contrast to VSG dimers and trimers which have a GPI-anchor on each subunit. There is experimental evidence that the signal for TfR concentration in the flagellar pocket is the number of GPI-anchors per molecule, a single anchor leading to a greater retention, and more than one anchor ready to escape to the surface of the cell body (Schwartz et al., 2005 [DOI](#); Tiengwe et al., 2017 [DOI](#)). These experiments manipulated the number of GPI-anchors on TfR, heterodimer with a single anchor on one subunit. Adding an anchor to the second subunit caused a change in localisation to the cell surface and endosome as judged by immunofluorescence. However, conditions that led to a few fold increase in expression of single GPI-anchored TfR, including transferrin shortage and growth at high cell density, resulted in the detection of TfR on the cell body surface (Mussmann et al., 2004 [DOI](#)). Whether the surface-localized TfR is functional remains unproven (Mussmann et al., 2004 [DOI](#); Schwartz et al., 2005 [DOI](#)).

TfR is encoded by a dispersed gene family with ~15 TfR variants. The experiments here started with a chance observation that one of these encoded a receptor with two GPI-anchors while the others have a single anchor. This was exploited to investigate how the presence of one or two GPI-anchors affected function. We show that the second GPI-anchor has no effect on the kinetics of uptake, or processing of transferrin through the endosome.

However, the second anchor resulted in ≥ 1.5 -fold increase in steady state receptor levels. Further, transferrin uptake experiments provided evidence that TfR was present over the entire cell surface and this was supported by immunofluorescence experiments. An analysis of the localisation of two other putatively GPI-anchored proteins showed that all were present over the entire cell surface. These experiments demonstrate that the trypanosome expresses a set of receptors over the entire cell surface inferring that there are effective mechanisms to resist antibody mediated recognition of conserved cell surface proteins.

Results

The BES7 VSG expression site encodes a transferrin receptor with two GPI-anchors

In *T. brucei*, the active VSG gene is transcribed from a bloodstream form expression site (BES) that is telomere proximal. BESs contain up to 12 other co-transcribed expression site associated genes (ESAGs) (Hertz-Fowler et al., 2008 [DOI](#); Pays et al., 2001 [DOI](#)). TfRs are heterodimers encoded by ESAG6 and ESAG7 (Chaudhri et al., 1994 [DOI](#); Salmon et al., 1994a [DOI](#)). The ESAG6 and 7 subunits share around 80% amino acid identity although ESAG6 has an additional 24 residues at the C-terminus preceding a GPI-anchor addition signal. Different BESs encode variants of ESAG6 and 7 with 90% or more amino acid sequence identity (Maier & Steverding, 2008 [DOI](#); Trevor et al., 2019 [DOI](#)). In *T. brucei* Lister 427, the isolate used in these experiments, there are 15 BESs (Barcons-Simon et al., 2023 [DOI](#)) and most were sequenced and annotated after transformation associated recombination cloning of telomeres (Hertz-Fowler et al., 2008 [DOI](#)). In the original annotation, BES7 stands out as it has two copies of ESAG6 but apparently lacked ESAG7 and as such would not be able to make a functional TfR. Here, a fresh analysis of the sequence was performed and indicated that the first ‘ESAG6’ gene annotated in BES7 actually encoded an ESAG7 with a C-terminal extension and a GPI-

anchor typical of an ESAG6. In addition, there was a second ESAG6 gene (ESAG6.2) downstream of the annotated copy (ESAG6.1), the two copies encode proteins with a single difference in 401 residues (M110→V). Phylogenetic analysis of ESAG6 and ESAG7 confirmed that a copy of ESAG7 is indeed present in BES7 along with a single copy of ESAG6 (Figure 1A [↗](#)).

The origin of the GPI-anchor attachment signal in BES7 ESAG7 was investigated by comparing the ESAG7 mRNA sequences from 13 different BESs. This showed that all the ESAG7 genes contained residual coding sequences for GPI-anchor addition but a set of small deletions between nucleotides 985 and 1068 in the ORF resulted in a frameshift leading to a premature stop codon at the C-terminal end effectively removing the GPI-anchor attachment sequence (Figure S1 [↗](#)). A comparison of BES7 ESAG6 and 7 showed that the C-terminal 120 amino acid sequences are identical and there was a clear prediction for a GPI-anchor on both using NetGPI 1.1 (Gislason et al., 2021 [↗](#)) (Figure S2 [↗](#)).

Cellular TfR protein levels are determined by BES identity and the number of GPI-anchors

ESAG6 and 7 protein levels were examined by western blotting in cell lines expressing VSGs and TfRs from three different BESs: BES3-VSG118, BES1-VSG221 and BES7-VSG224 (Figure 1B [↗](#)). TfRs have several N-linked glycosylation sites and whole cell lysates were treated with PNGaseF to simplify analysis. First, in BES7 cells there was a shift in the apparent molecular weight of ESAG7 from ~30 kDa to ~38 kDa, which is consistent with the addition of a GPI-anchor (Figure 1B [↗](#)), such a change in mobility on the addition of a GPI-anchor to ESAG7 has been reported previously (Tiengwe et al., 2017 [↗](#)). Second, the identity of the active BES affected both ESAG6 and ESAG7 protein levels, there were increasing levels of expression: BES3 < BES1 < BES7 (Figure 1B [↗](#)). Lastly, when the BES7 cell line was modified to exchange VSG224 to VSG221 (see below), this had little effect on TfR expression, suggesting the BES identity was the dominant determinant rather than the VSG type. The rabbit TfR antiserum used was raised against the TfR from BES1 (Gerrits et al., 2002 [↗](#)) and is the same as used in most other studies (Mussmann et al., 2003 [↗](#); Tiengwe et al., 2017 [↗](#)). The amino acid sequence identity between BES2 TfR and BES1 TfR is >98% and BES7 TfR is >92% and it is unlikely that the difference in TfR levels result from differential reactivity of the serum to different TfRs. Support for the different levels of TfR expression was obtained when the levels of TfR mRNAs were estimated in the BES3, BES1 and BES7 cell lines using RNAseq (Table S1 [↗](#)). The level of TfR mRNAs were expressed as a ratio (ESAG6+ESAG7):VSG and were consistent with the differences seen in protein levels (BES3, 0.022 < BES1, 0.038 < BES7, 0.077).

We next investigated whether GPI-anchor number contributed to the variation in cellular TfR levels by making a set of transgenic cell lines (Figure 2A [↗](#)). First, VSG224 in BES7 was changed to VSG221. Next, the cell line expressing VSG221 from BES7 was modified to alter the ESAG7 by exchanging the 3' end of the ORF with that from BES1 ESAG7 resulting in the loss of the GPI-anchor signal. As a control, a second cell line in which the ESAG7 gene retained the GPI-anchor addition sequence was made. Similar modifications were performed on cells expressing VSG221 from BES1, cell lines expressing either BES1 ESAG7 modified with the C-terminal GPI-anchor sequence from BES7 ESAG7 or the endogenous BES1 ESAG7 were made. The result was four cell lines: all expressed VSG221 from either BES1 or BES7 and TfR with or without a GPI-anchor on ESAG7.

The cell lines were characterised by western blotting of PNGaseF-treated whole cell lysates (Figure 2B [↗](#)). The addition of a GPI-anchor signal sequence to ESAG7 resulted in an increase in both ESAG6 and 7. To estimate this change, whole cell lysates were minimally separated by SDS-PAGE to collapse the ESAG6 and 7 proteins into a single band prior to western blotting. Relative quantitation was obtained using a titration curve of the highest TfR expresser and normalised against SCD6 for loading (Figure 2C [↗](#)). Three biological replicates were performed and the levels of both TfRs from BES1 is significantly lower ($p < 0.01$), ~3.5 fold, compared to BES7, regardless of GPI-anchor number. The presence of a GPI-anchor on ESAG7, leading to a double-anchored TfR, significantly increased ($p < 0.05$) the expression level by approximately 1.9-fold in BES1 and about 1.5-fold in BES7. This provides strong evidence that the presence of a GPI-anchor on ESAG7

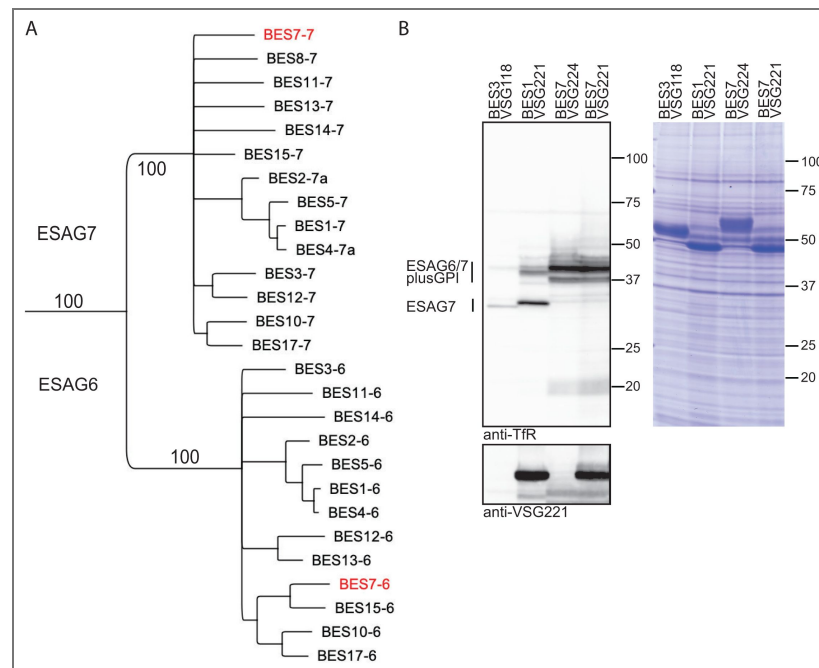


Figure 1. *T. brucei* transferrin receptor expressed from BES7 contains GPI-anchored ESAG7.

(A) Phylogenetic tree showing the relationship between ESAG6 and ESAG7 proteins from *T. brucei* Lister 427 (Hertz-Fowler et al., 2008). The tree was assembled in Geneious Prime (www.geneious.com) using an alignment by Muscle 5.1 followed by a neighbour joining tree with 100 bootstraps and no outgroup on residues 1 to 332 of the alignment. BES7 ESAG 6.1 and 7 are shown in red. (B) Variation in transferrin receptor expression in cell lines expressing VSGs from different bloodstream form expression sites (BES). BES3-VSG118, BES1-VSG221, BES7-VSG224 cell lines are unmanipulated and the BES7-VSG221 was made after exchanging VSG224 for VSG221 (see Figure 2A). Cell lysates were treated with PNGaseF and 2×10^6 cells equivalents were loaded. Western blots were probed with anti-TfR, and then overprobed with anti-VSG221. ESAG6 and ESAG7 plus GPI-anchor variants are indicated; BES3-VSG118 and BES1-VSG221 cell lysates contain ESAG7 without the GPI anchor. BES3-VSG118 and BES7-VSG224 cell lysates did not show any VSG221. The presence of VSG221 protein confirmed the switching of BES7 VSG224 to VSG221. Switching in VSG type did not affect the expression pattern of native TfR (ESAG6/7 plus GPI) protein. Coomassie blue stained SDS-PAGE gel to demonstrate loading of samples.

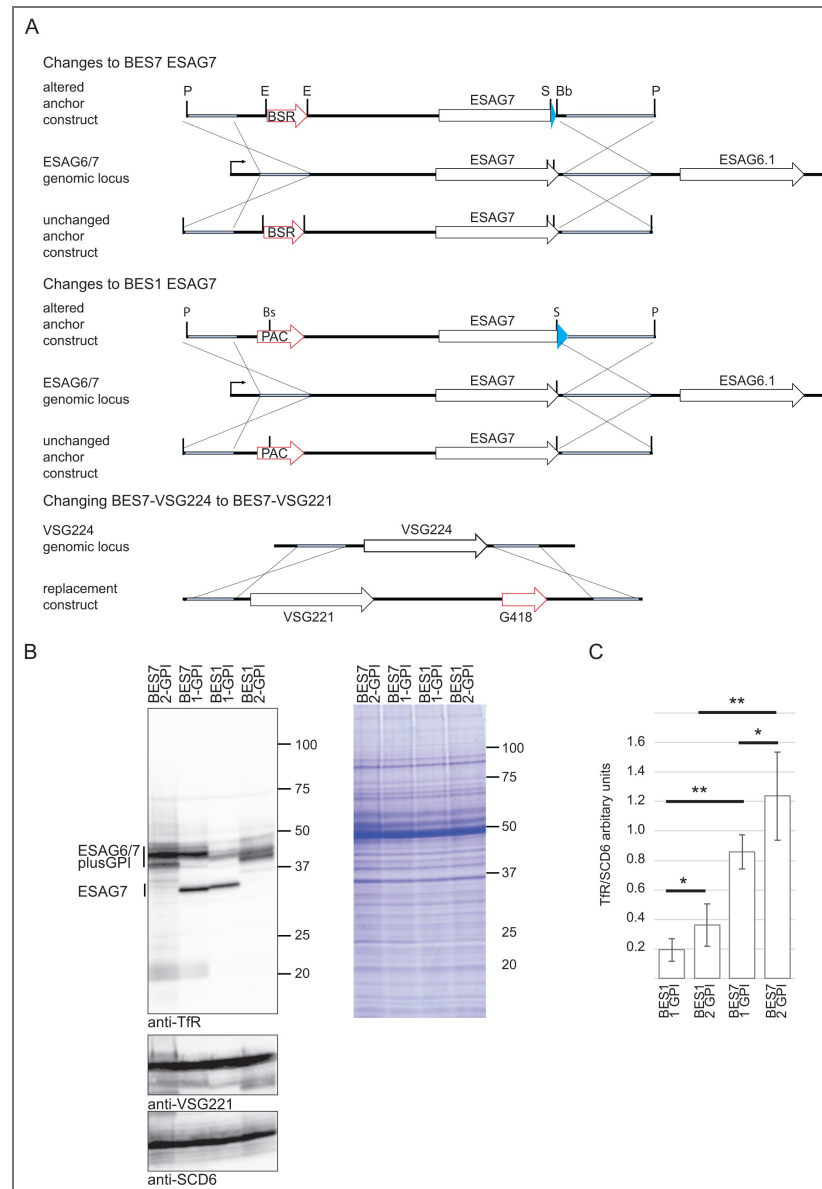


Figure 2. The effect of BES identity and the number of GPI-anchors on the cellular Tfr levels.

(A) Schematic illustrating the production of cell lines. For the addition and removal of ESAG7 GPI-anchor addition sequences, two constructs were used for each of BES1 and BES7. A description of the construct is provided in the methods. For BES7, the same process was used, one construct removing the GPI-anchor from ESAG7, the other leaving it intact. For BES1, integration of one resulted in modification of ESAG7 to add a GPI-anchor, integration of the second resulted in an unchanged ESAG7 polypeptide sequence. Restriction enzyme sites used in cloning: P, PacI; E, EcoRI; S, SacI; Bs, BstEII and Bb, BsrBI. The VSG in BES7-VSG224 cells was switched to VSG221 to remove any VSG-dependent variation. The targeting construct conferred G418 resistance. (B) Transferrin receptor expression in the following cell lines: BES7-VSG221 single (1-GPI) and double (2-GPI) GPI-anchored Tfr and BES1-VSG221 single and double GPI-anchored Tfr. Cell lysates were treated with PNGaseF and 2×10^6 cells equivalents were loaded per gel track. A western blot of the samples was probed with anti-Tfr then over probed with anti-VSG221 and anti-SCD6. ESAG6 and ESAG7 plus GPI-anchor variants are indicated. (C) Tfr quantitation estimates from a minimum of three biological replicates of each cell line, error bars display the standard deviation for each sample. Samples plus a dilution series from BES7-221 were minimally separated by SDS-PAGE to compress Tfr components to a single band for analysis after western blotting, probed with anti-Tfr and anti-SCD6. Donkey anti-rabbit Alexa 488 was used as secondary antibody and band intensity was calculated by ImageJ ensuring that signals taken for each reading were below saturation levels. Relative quantitation was determined by mapping values to those obtained from a dilution series. Measurements for SCD6 were used to normalise cell loading with the value for BES1-1GPI anchor samples given an arbitrary value of 1. Statistical analysis was used a two-tailed paired t-test, $**p \leq 0.01$; $*p \leq 0.05$.

contributes to the higher cellular expression level of TfR, which could result from increased protein stability, reported for VSGs, (Bangs et al., 1986 [↗](#); Engstler et al., 2004 [↗](#)) or a higher rate of turnover and/or greater shedding of the single GPI-anchored TfR (Schwartz et al. 2005 [↗](#)).

Transferrin uptake kinetics are similar in cells expressing TfRs with either one or two GPI-anchors

To investigate whether anchor number influenced function, BES7-VSG221 cell lines expressing single or double GPI-anchored TfR were used. It has already been shown that a TfR with double GPI-anchors is functional (Tiengwe et al., 2017 [↗](#)), here the kinetics of transferrin binding and uptake by single and double GPI-anchor TfRs were compared. The experimental approach was similar to that developed to characterise the binding and endocytosis of VSG antibodies (Engstler et al., 2007 [↗](#)). The medium used to culture trypanosomes contained ~2.5 µM transferrin (Kakuta et al., 1997 [↗](#)), Alexa 568 labelled bovine holo-transferrin (TfA568) was added to cultures, without any other manipulation, to a final concentration of 0.5 µM (so that 1 in 6 molecules of Tf in the medium is now fluorescent) and cells were then fixed over a time course from 15 to 600 seconds and imaged by epifluorescence microscopy (Figure 3A [↗](#) and S3A). The on rate for binding of Tf to TfR is very rapid ($4.5 \times 10^5 \text{ M}^{-1}\text{s}^{-1}$ at pH 7.4 for BES1 TfR and bovine Tf) (Trevor et al., 2019 [↗](#)) and binding will occur to unoccupied receptor almost immediately. At 15 seconds there was variation between individual cells with TfA568 present on the flagellum, across the cell body and the flagellar pocket in both cell lines. There was also variation at 60 seconds, TfA568 was present in the endosomal compartment of most cells, and the cell surface in some cells. At 600 seconds the TfA568 was predominantly within the endosomal compartment. The variation was also examined by quantifying the fluorescence from individual cells (Figure 3B [↗](#)), this showed the range of variation and an increase in fluorescence between 15 and 60 seconds followed by a decrease by 600 seconds. Fluorescence intensity was also determined across transverse sections of the cell (Figure 3C [↗](#)), this showed some cells with internal signal and others with two peaks consistent with a cell surface localisation. There was a stronger signal from cells expressing TfR with two GPI-anchors, consistent with higher expression levels, but no consistent differences in distribution were seen.

The same TfA568 uptake assay was performed with the BES1-VSG221 cells expressing TfRs with one or two GPI-anchors (Figure 3D [↗](#) and S3B). There was a reduced binding and uptake of TfA568 in BES1 cells compared to BES7 cells as might be expected given lower levels of the receptor. Otherwise, the time course of uptake was similar: at 15 seconds TfA568 was present on the flagellum and cell body surface as well as the flagellar pocket in both cell lines, at 60 seconds TfA568 on the cell surface was reduced and some was now present in the flagellar pocket and endosomal compartments, and by 600 seconds TfA568 was substantially within endosomal compartments. In both BES7 and BES1 cell lines, fluorescence distribution patterns across the cells showed variability at 15 and 60 sec (Figure 3E-F [↗](#)), but appeared more uniform at 600 sec.

Previous studies have shown that some TfRs do not bind canine transferrin (Mussmann et al., 2003 [↗](#)). Canine transferrin has 501/686 amino acid sequence identities with bovine transferrin (Figure S4A [↗](#)) and it was used as a negative control in the same transferrin binding and uptake assay above using BES7-VSG221 cells expressing TfR with either one or two GPI-anchors (Figure S4B [↗](#)). No cell binding or uptake of canine Tf in either cell line was observed. In addition, a competition assay for binding of fluorescent Tf was performed by adding 5- or 10-fold excess of unlabelled Tf at the same time (Figure S4C [↗](#)). The signal from labelled Tf was reduced by the competition indicating saturable binding, consistent with a receptor mediated process. These findings provide evidence that the binding of bovine transferrin to the cell surface is receptor dependent. An important point is that fluorescent Tf was added to cells in culture without any other manipulation as preliminary experiments indicated that centrifugation, washing and resuspension in medium led to a reduction in cellular TfR for up to 2 hours (Figure S4D [↗](#)).

Together, these experiments support a model in which the TfR is present on the cell surface, and endocytosis is a rapid process. The greater Tf binding and uptake in BES7 compared to BES1 cells provides evidence that TfR abundance limits the amount of Tf endocytosed. The number of GPI-

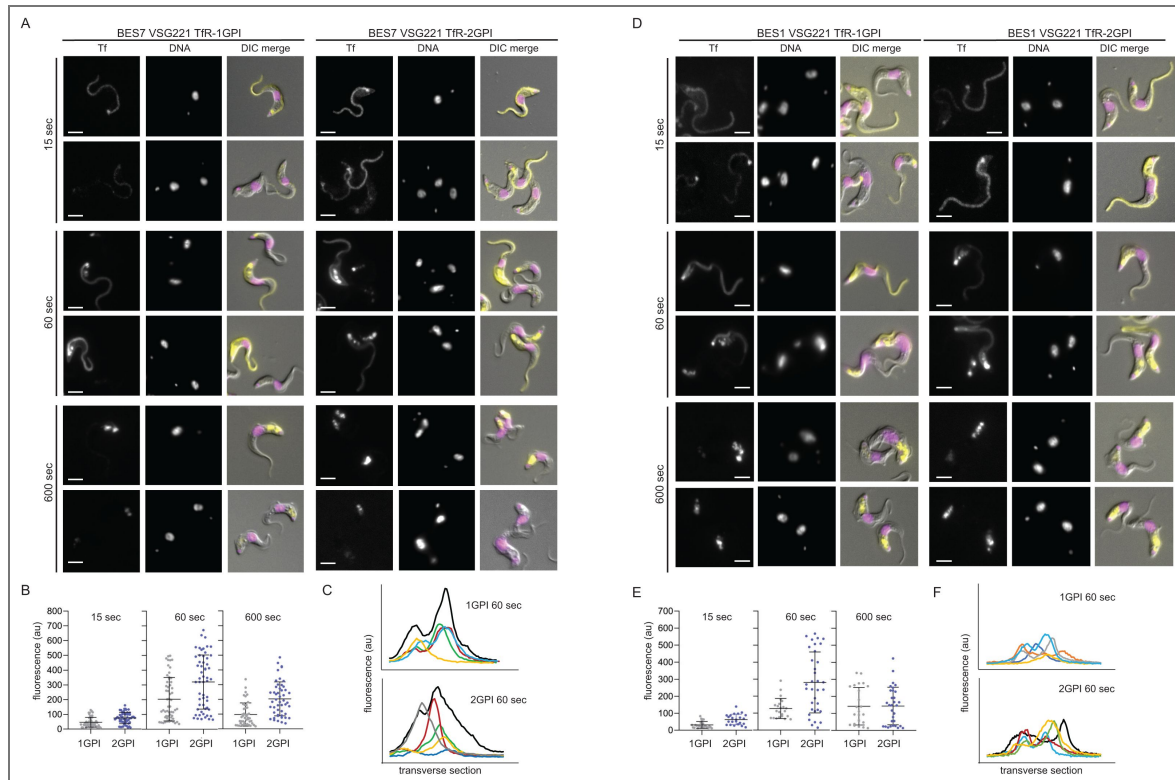


Figure 3. Transferrin endocytosis is rapid, in both single and double GPI-anchored cells.

Initial binding is to the cell surface and internalization is detected within 60 sec. Cultures of single (1-GPI) or double (2-GPI) GPI-anchored TfR expressing BES7-VSG221 and BES1-VSG221 cells were supplemented with 500 nM Alexa568 labelled holo-transferrin (TfA568) and incubated for 15 sec, 60 sec or 600 sec and fixed by addition of formaldehyde for 5 min. Following washing with PBS containing 1% bovine serum albumin cells were visualized under Zeiss Axio Imager.Z2 with X100 objective. Images in all these experimental time points for both 1- and 2GPI TfR cells were captured with same intensity of excitation and exposure time. (A) Images showing the location of bound or endocytosed transferrin in representative cells in populations and at time points as indicated. Scale bar is 5 μ m. Images were processed with ZEN Blue 3.4 software without any deconvolution. Larger numbers of cells are shown in [Supplementary Figure 3](#). (B) The variation between individual cells in transferrin binding and uptake and endocytosis shown by measurements of total fluorescence intensity of TfA568 in BES7 VSG221 1- and 2-GPI TfR cells at 15 sec, 60 sec and 600 sec. Measurements were obtained using Zen Blue 3.4 software and normalizing the background fluorescence ([Table S2](#)). Scatter plots of measurements of individual cells are shown for 1-GPI (grey) and 2-GPI (purple) TfRs. The average and standard deviation are shown for each cell line. (C) TfA568 fluorescence intensity distribution across transverse sections of randomly selected cells was measured using ImageJ software. The plots are shown for both BES7 1- and 2-GPI cell lines, showing the pattern of individual cells in different colours at 60 sec and 600 sec. (D) (E) and (F) Same as (A) (B) and (C) but with BES1-VSG221 1- and 2GPI TfR expressing cells and [Table S3](#) corresponds to (E).

anchors on the TfR has no obvious influence on Tf uptake but does affect receptor levels.

The finding that TfA568 was bound over the whole cell surface at early time points in the uptake assays suggested a similar localisation for the TfR. This is not consistent with some previous studies that found TfR and other GPI-anchored receptors sequestered in the flagellar pocket (Duncan et al., 2024 [\[1\]](#); Mussmann et al., 2004 [\[2\]](#); Steverding et al., 1994a [\[3\]](#), 1995 [\[4\]](#); Tiengwe et al., 2017 [\[5\]](#); Vanhollebeke et al., 2008 [\[6\]](#)) or that expression level determined localisation (Mussmann et al., 2003 [\[7\]](#)). We went on to characterise the cellular localisation of single and double GPI-anchored TfR along with two other GPI-anchored receptors.

Localisation of the transferrin receptor by immunofluorescence

Previous reports provided evidence that in BES1-VSG221 cells grown in the presence of 10% FBS, the cell surface TfR is located or concentrated within the flagellar pocket and endosome, rather than dispersed on the cell body (Ligtenberg et al., 1994 [\[8\]](#); Mussmann et al., 2003 [\[7\]](#); Salmon et al., 1994a [\[9\]](#); Tiengwe et al., 2017 [\[5\]](#)). In contrast, when cells are treated to increase TfR expression by roughly three to fivefold, for example when grown to high density or in canine serum, TfR localizes on the cell body (Mussmann et al., 2003 [\[7\]](#)). To resolve this discrepancy, the localisation of the TfR was investigated using immunofluorescence. Based on the transferrin uptake experiments, we reasoned that the localisation of the TfR would be very dynamic and thus chose to fix the cells in culture without any centrifugation or washing, this approach was supported by preliminary experiments comparing protocols (Figure S5A-B [\[10\]](#)) that showed both the composition of the fixative and absence of a washing step altered the outcome.

Four cell lines, BES1-VSG221 and BES7-VSG221 cells each expressing TfR with one or two GPI-anchors, were investigated. Fixative was added to cells in culture without any other manipulations, the final concentrations were 4% formaldehyde and 0.2% glutaraldehyde and TfR localisation was observed by immunofluorescence after settling cells on microscope slides. Under these experimental conditions, the TfR in all four cell lines had a distribution across the entire cell surface including the flagellar pocket and in internal compartments in some cells (Figure 4A [\[11\]](#)). Fluorescence intensity measurements across transverse sections of both the BES7 1- and 2 GPI cells confirmed that TfR is indeed present on the surface of some cells as seen by two peaks in fluorescence corresponding to the cell periphery in some cells (Figure 4B [\[12\]](#)). As in the Tf uptake experiments, the strength of signal as measured by total fluorescence intensity from individual cells showed that TfR abundance varied between individual cells in populations (Figure 4C [\[13\]](#) and Figure S5C [\[10\]](#)). These measurements also showed that the presence of two GPI-anchors on TfR increased levels by about 1.5-fold compared to the single anchored receptor and that there is a ~3.5-fold higher TfR expression in BES7 compared to BES1 cells regardless of GPI-anchor number. These measurements support the western blot quantitation (Figure 2 [\[14\]](#)) and also provide evidence that the immunofluorescence signal does not contain a significant level of non-specific background.

Structured Illumination Microscopy (SIM) was used to obtain more information on TfR localisation in the same four cell lines as above (Figure 5 [\[15\]](#)). TfR was present on the cell surface including the flagellar pocket, cell body and flagellum, and in the endosome (single stack images) (Figure 5A [\[16\]](#)). Three-dimensional reconstruction of the localization of TfR confirmed that it is distributed across the cell surface and in endosomes (Figure 5B [\[17\]](#)).

These experiments collectively provide strong evidence that TfR is indeed present on the *T. brucei* cell surface under normal growth conditions and that the localization pattern is independent of the presence or absence of a GPI-anchor in the ESAG7 subunit.

Cell surface localisation is shared by two other GPI-anchored receptors

The finding that TfR is distributed over the whole cell surface led to an investigation of other receptors. Two were selected on the basis of being predicted to have one or more GPI-anchors and a non-expressing negative control being available. The first was the haptoglobin haemoglobin

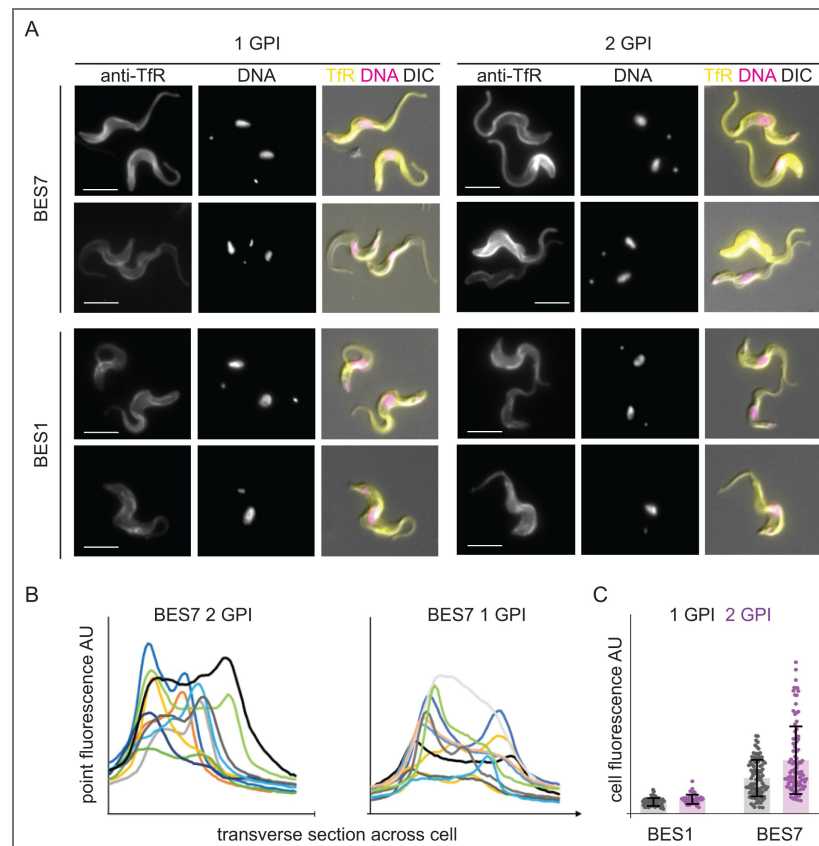


Figure 4. Immunofluorescence localisation of single and double GPI-anchored transferrin receptors to the entire plasma membrane.

Cell lines expressing VSG221 and either 1- or 2-GPI TfR variants were fixed in culture with final concentrations of 4% formaldehyde and 0.2% glutaraldehyde. Immunofluorescence localisations were performed with rabbit anti-TfR followed by donkey anti-rabbit Alexa 488 (green). Cells were visualized using a Zeiss Axio Imager.Z2 with X100 objective. Images in all these experimental sets were captured with same intensity of excitation and exposure time. (A) Localization in cell lines as indicated. Scale bar is 5 µm. Images were processed with ZEN Blue 3.4 software. A larger number of cells is shown in [Supp. Figure 5](#). (B) Fluorescence distribution across transverse sections of the cells was measured using ImageJ software. Plots are shown for both BES7-VSG221 TfR cell lines, each cell in a different colour. (C) Mean anti-TfR fluorescence intensity in individual cells of both BES1 VSG221 and BES7 VSG221 1- and 2-GPI TfR cell population was measured using ZEN Blue 3.4 and normalized against the background fluorescence ([Table S4](#)). Bar diagram shows the fluorescence intensity of either 1-GPI (grey) or 2-GPI (purple) TfR from individual cells. The columns represent the average and the error bars show the standard deviation.

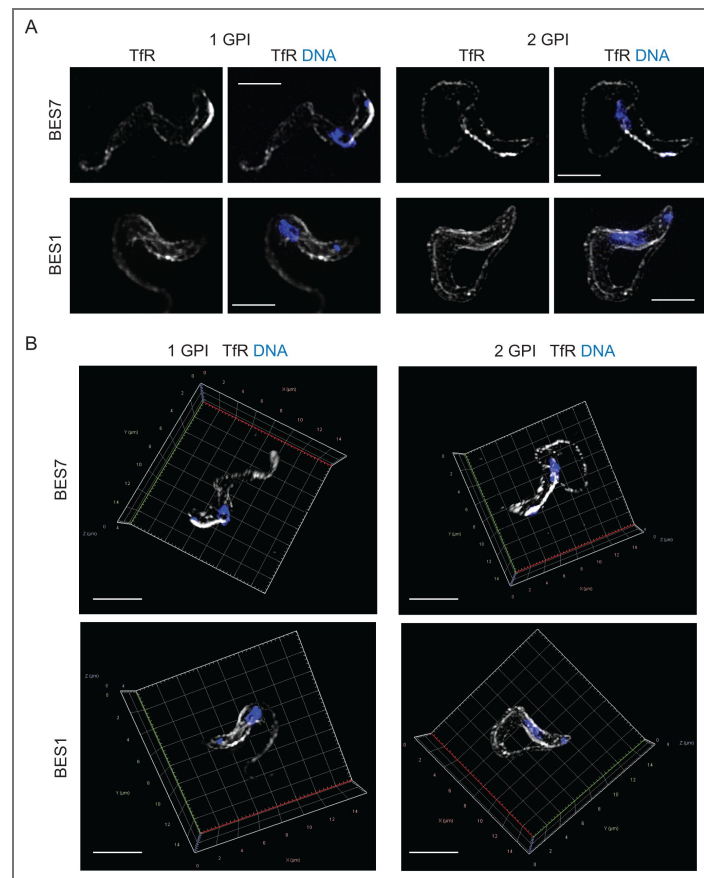


Figure 5. Localization of single and double GPI-anchored transferrin receptors under 3D structured illumination microscope.

Cells expressing either 1- or 2-GPI-anchored TfR were fixed in culture with final concentrations of 4% formaldehyde and 0.2% glutaraldehyde. Immunofluorescence localisation was performed with rabbit anti-TfR and donkey anti-rabbit Alexa 488. Cells were visualized using a Zeiss Elyra7 microscope with X63 objective. Images in all these experimental sets were captured with same intensity of excitation and exposure time. (A) Localization of TfR in different cell lines as indicated. Scale bar is 5 μm . (B) 3D projection generated using ZEN Blue 3.4 software example cells from the cell lines as indicated. These images show the localization pattern of TfR on the cell surface as well as within intracellular compartments. X-Y-Z axis is indicated. Scale bar 5 μm .

receptor (HpHbR) for which $+/+$ and $-/-$ cell lines were used. The structure of most of HpHbR is known and it is a monomer in solution but two or more receptors can bind a single ligand (Higgins et al., 2013 [↗](#); Lane-Serff et al., 2014 [↗](#)). HpHbR has a low copy number, 300 to 400 per cell (Vanhollebeke et al., 2008 [↗](#)) and, in the absence of a suitable antibody, a ligand uptake assay was used. The second receptor analysed was the complement factor H receptor (FHR), this was used after a fortuitous observation that although it is not expressed when cells are grown in medium containing 10% foetal bovine serum (Macleod et al., 2020 [↗](#)), it is expressed at high levels when the same cells are grown in medium with 10% rabbit serum replacing the foetal bovine serum. The structure of FHR is known and it is a monomer with a single GPI-anchor (Macleod et al., 2020 [↗](#)).

Alexa 568-labelled HpHb (HpHbA568) was used in uptake assays with HpHbR $+/+$ and $-/-$ cells as for transferrin uptake (Figure 6A [↗](#)). There was binding of ligand to the cell surface by 15 seconds, uptake into the endosome by 60 seconds, and a location consistent with the lysosome by 600 seconds. Binding and uptake were not detected in HpHbR $-/-$ cells (Figure 6A [↗](#)). The kinetics of uptake were similar to those for transferrin. Presence of HpHbA568 on the cell surface was confirmed by the fluorescence signal measurement across the cells at 15 seconds, which showed two distinct peaks consistent with peripheral localisation for some cells (Figure 6B [↗](#)), the noise in the quantification is due to the low copy number of the receptor. Again, there was variability in binding and uptake of HpHb across the population (Figure S6 [↗](#)).

In trypanosomes cultured in medium containing 10% rabbit serum, expression of the FHR was readily detected whereas in medium containing foetal bovine serum (FBS) it is not (Figure 7A [↗](#)). Cells cultured with rabbit serum were used to determine the localisation of FHR by immunofluorescence following the same procedure used for the transferrin receptor (Figure 7B [↗](#)). The FHR was clearly localised over the entire cell surface. Two controls for specificity of antibody binding were performed: first cells were grown in medium supplemented with foetal bovine serum and second with a no primary antibody control for the cells grown with rabbit serum to control for cell surface immunoglobulin originating from the growth medium. Neither control gave a signal (Figure S7 [↗](#)). SIM analysis further provided spatial resolution along with reconfirming the distribution of the FHR across the trypanosome cell surface and flagellum as well as within the flagellar pocket (Figure 7C [↗](#)).

These experiments show that the detection of TfR on the cell surface is part of a general pattern where GPI-anchored receptors are distributed across the cell body plasma membrane.

Discussion

Any pathogen that has evolved to establish and maintain a long-term infection within a mammalian host must be able to acquire nutrients as well as be resistant to both the innate and adaptive immune responses of the host. African trypanosomes have evolved a strategy that is dependent on variant surface glycoprotein (VSG), a superabundant protein that forms a densely packed coat on the cell surface. Low frequency switching to a different VSG allows the population to persist (Mugnier et al., 2015 [↗](#)). The VSG coat appears to prevent antibodies, and possibly innate immunity factors, gaining immediate access to the plasma membrane (Schwede et al., 2011 [↗](#)). However, such a coat cannot be totally impermeable as trypanosomes require host macromolecules such as transferrin and haptoglobin-haemoglobin which they acquire through receptor mediated endocytosis (Lane-Serff et al., 2014 [↗](#); Salmon et al., 1994b [↗](#); Steverding et al., 1994b [↗](#), 1995 [↗](#); Trevor et al., 2019 [↗](#); Vanhollebeke et al., 2008 [↗](#)).

Why these receptors do not result in the trypanosomes becoming susceptible to immune recognition and attack is central to understanding the mechanisms of infection (Banerjee et al., 2024 [↗](#)). One model proposed that the receptors are sequestered in the flagellar pocket, an invagination of the plasma membrane at the base of the flagellum, and are not accessible to immune effectors. The idea that sequestration in the flagellar pocket allowed receptors to hide from host immune effectors was first challenged by the discovery of invariant surface glycoproteins (ISGs) (Ziegelbauer et al., 1992 [↗](#); Ziegelbauer & Overath, 1992 [↗](#)) that were shown

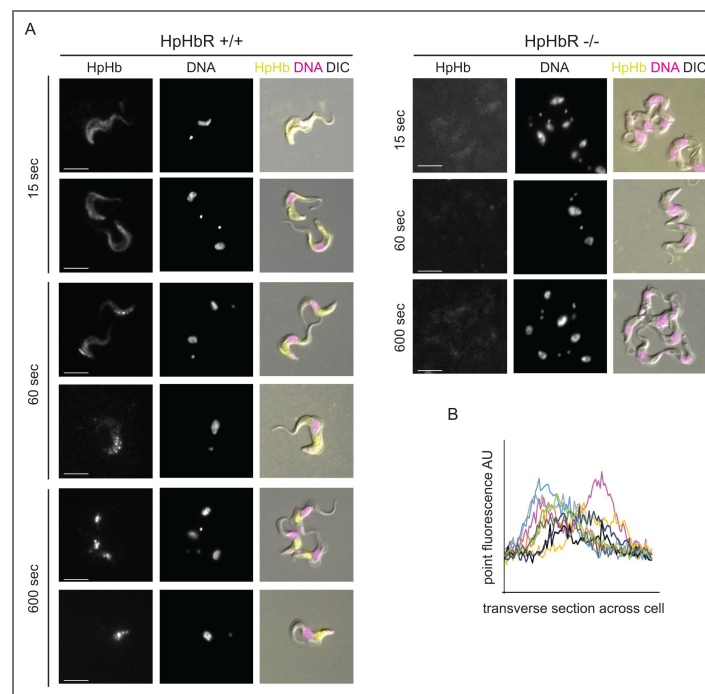


Figure 6. Haptoglobin-haemoglobin endocytosis assay.

Initial binding is to the cell surface followed by rapid endocytosis. Cells were supplemented with 500 nM Alexa 568 labelled HpHb, incubated for 15 sec, 60 sec or 600 sec and fixed in culture with 1% formaldehyde. Following washing with PBS containing 1% bovine serum albumin. Cells were visualized under Zeiss Axio Imager.Z2 with x100 objective. Images in all these experimental time points were captured with same excitation intensity and exposure time. (A) Images showing the location of HpHb in *+/+* and *-/-* cells at different time points in representative cells as indicated. Images were processed with ZEN Blue 3.4 software. Scale bar is 5 μ m. A greater number of cells is shown in [Supp. Figure 6](#). (B) HpHb fluorescence intensity distribution over transverse section of cells at 15 sec measured using ImageJ software. Individual cells are shown in different colours.

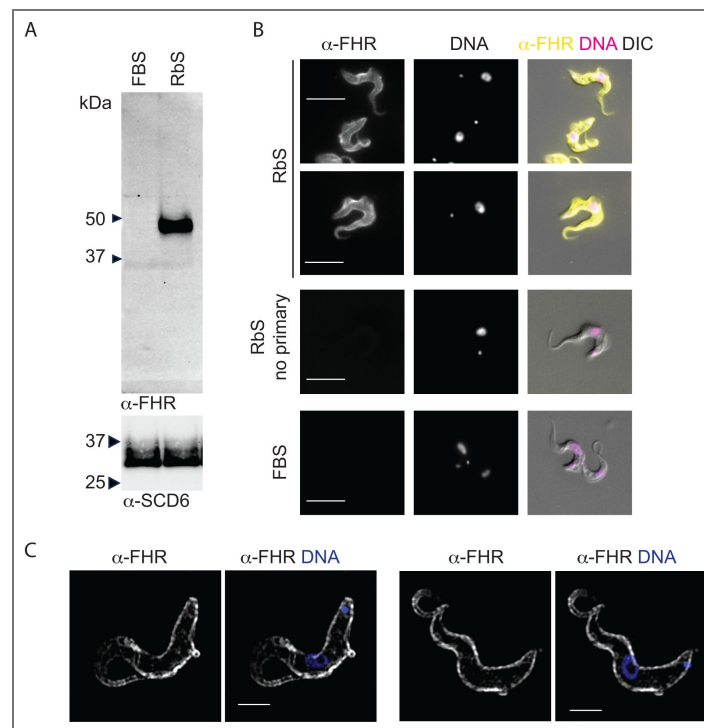


Figure 7. The complement factor H receptor localises to the cell surface.

The observation that abundant FHR expression occurred in cells grown in rabbit serum but not in FBS supplemented medium was used to investigate its localisation. (A) Western blot analysis of FHR expression in *T. brucei* EATRO1125 cells grown in medium supplemented with either foetal bovine serum or rabbit serum. Cell lysates were not treated with PNGaseF and 2×10^6 cells equivalent were loaded. The blot was probed with rabbit anti-FHR and then overprobed with anti-SCD6 for loading. (B) Immunofluorescence determination of localisation of FHR in cell lines as indicated. Cells were fixed in culture with final concentration of 4% formaldehyde and 0.2% glutaraldehyde. Rabbit anti-FHR antibody was followed by donkey anti-rabbit Alexa 488 secondary antibody. A set of cells grown in RbS, fixed similarly and stained only with secondary antibody was used as a control for residual rabbit IgG from the growth medium and cells grown in FBS for non-expressers. Cells were visualized using a Zeiss Axio Imager.Z2 with x100 objective. Scale bar is 5 μ m. A larger number of cells is shown in [Supp. Figure 7](#). (C) Localisation of FHR by 3D structured illumination microscopy. Cells fixed as above were visualised using a Zeiss Elyra7 with x63 objective. Images were processed with ZEN Blue 3.4 software. Two representative cells images showing FHR in white and merged panel with DNA in blue. Scale bar is 5 μ m.

to be distributed over the whole cell surface (Macleod et al., 2022 [↗](#); Ziegelbauer & Overath, 1993 [↗](#)). Two ISGs were subsequently identified as receptors, one for complement C3 and another for the Fc region of immunoglobulins (Macleod et al., 2022 [↗](#); Mikkelsen et al., 2024 [↗](#)). Both the ability to access ligands and the structures of the extracellular domains of both ISGs indicated that they were unlikely to be physically shielded by the VSG coat. The ISGs are type 1 transmembrane proteins with a large extracellular domain, a single transmembrane pass and a small cytoplasmic tail. In contrast, VSGs have two or more GPI-anchors whereas the two characterised nutrient receptors have a single GPI-anchor. Could this difference in GPI-anchor number result in selective retention of nutrient receptors in the flagellar pocket (Tiengwe et al., 2017 [↗](#)) as it is difficult to envisage a requirement for any amino acid sequence motifs in the context of variation of VSGs.

Here, we have investigated the distribution of GPI-anchored receptors on the cell surface of mammalian infective *Trypanosoma brucei*. The main findings are: (1) The majority of TfRs encoded in the genome have a single GPI-anchor but there is at least one with two GPI-anchors. (2) The single anchored TfRs probably arose from loss of one anchor as opposed to gain in the double anchored version. The loss of the GPI-anchor from ESAG7 may provide a selective advantage possibly reducing exposure to the host adaptive immune system without affecting transferrin acquisition (Trevor et al., 2019 [↗](#)). (3) The steady state level of TfR varies with the identity of the transcribed BES. (4) Single or double anchored TfRs bind transferrin, as previously reported (Tiengwe et al., 2017 [↗](#)) and result in similar kinetics of transferrin uptake, with movement from the cell surface to the endosome occurring within two minutes. The rapid kinetics of uptake indicates the receptor ligand complex is probably subject to the same hydrodynamic forces as antibody VSG complexes (Engstler et al., 2007 [↗](#)) resulting in a directional movement towards the posterior end of the cell, the location of the flagellar pocket and endocytosis. (5) Adding a second GPI-anchor to a TfR resulted in a ~1.5-fold increase in cellular receptor levels. (6) Both single and double GPI-anchored TfRs are present over the whole cell surface as seen by both transferrin and antibody binding. (7) Two other GPI-anchored receptors, the HpHbR and FHR, both probably with a single GPI-anchor, are present over the whole cell surface.

The earlier findings that the TfR, and subsequently the HpHbR, are localised to the flagellar pocket is an attractive model as it provided an easy explanation for why the receptors did not result in immune attack. The experiments presented here do not agree with this model and it is worth discussing the reasons why these experimental differences have arisen. Here the cells were fixed in culture using formaldehyde and a higher concentration (0.2%) of glutaraldehyde than in previous reports. In contrast, much of the previous work involved manipulations of the trypanosomes before determining either transferrin binding or TfR/ HpHbR localisation by immunofluorescence. The most common is centrifugation and washing prior to fixation for immunofluorescence (Engstler et al., 2004 [↗](#); Roggy & Bangs, 1999 [↗](#); Tiengwe et al., 2017 [↗](#); Vanhollebeke et al., 2008 [↗](#)), a process that is slow when compared to the very rapid endocytosis of surface bound transferrin (Figure 3 [↗](#)) and providing time for the ligand bound receptors to be cleared from the cell surface. This would be even more apparent if the experimental manipulation reduced the export of new or recycled TfRs to the cell surface. An example of how protocols determine outcomes is the variation in the measured half-life of TfR, which ranges from ~40 min to 7 hrs depending on the different experimental protocols used (Biebinger et al., 2003 [↗](#); Kabiri & Steverding, 2000 [↗](#); Mussmann et al., 2004 [↗](#); Tiengwe et al., 2017 [↗](#)).

The experiments involving transferring BES1-VSG221 cells to medium in which FBS had been replaced by canine serum (Mussmann et al., 2003 [↗](#)) are informative. The BES1 TfR does not bind canine transferrin and cell proliferation arrested. After ~5 days, proliferating cells appeared in the culture having switched BES and thus TfR orthologue to one able to bind canine transferrin (van Luenen et al., 2005 [↗](#)). In addition, supplementing the canine serum based medium with bovine transferrin prevented arrest of proliferation indicating that it was due to lack of transferrin alone (Mussmann et al., 2004 [↗](#)). In contrast to experiments that wash cells into solutions that do not support growth, the transfer of BES1-VSG221 cells to medium with canine serum starves the cells of just transferrin. On transfer from FBS to canine serum, the TfR abundance increased 4 to 5-fold and the receptor was present all over the cell surface (Mussmann

et al., 2003 [\[1\]](#)). It is unlikely that the distribution of TfR on the cell surface results from a retention system being overwhelmed given the variation in TfR expression observed from different BESs (Figure 1 [\[1\]](#)) but is due to an absence of ligand-mediated uptake. Different TfRs have a wide range of affinities for transferrins (KD from as low as 1.4 nM to >5 µM) (Trevor et al., 2019 [\[2\]](#)) and there will be an equilibrium with the relative amounts of TfR on the cell body surface compared with the flagellar pocket and internal structures that will vary with the identity of the TfR and the identity and concentration of transferrin.

The experiments here were informed by the kinetics of rapid transferrin uptake (Figure 3 [\[1\]](#) and as an unpublished observation in (Mussmann et al., 2004 [\[3\]](#)) and we did not manipulate the cells before ligand uptake experiments or prior to fixation for immunolocalization. The result was clear evidence that TfR is expressed over the whole cell surface and is not restricted to the flagellar pocket. This was reinforced by the localisation of the HpHbR and FHR, both of which are monomeric (Lane-Serff et al., 2014 [\[4\]](#); Macleod et al., 2020 [\[5\]](#)), contain a single GPI-anchor, and are also present over the whole cell surface. We also observed variation in TfR and HpHbR abundance between individual cells, as has been seen previously for TfR (Duncan et al., 2024 [\[6\]](#); Steverding et al., 1994b [\[7\]](#)), and this suggests sensitivity to the requirements at the individual cell level. Whether this relates to cell cycle stage or any developmental progression remains to be investigated. The expression of the TfR is sensitive to the availability of Fe³⁺ with a several fold increases in mRNA and protein induced by iron chelators (Gilbert Carbajo et al., 2021 [\[8\]](#)). Growth in medium dependent on rabbit serum resulted in high levels of expression of the FHR, this could result from a factor in rabbit serum inducing expression or failure to bind ligand.

The findings here still leave the question of how the trypanosome avoids receptor antibody-based clearance by the host immune system? Vaccination of mice with ISGs did not result in any protection (Ziegelbauer & Overath, 1993 [\[9\]](#)), this shows that receptors do not have to be sequestered in the flagellar pocket to avoid the host adaptive immune system. The effective killing of trypanosomes by antibody-drug conjugates recognising the HpHbR shows that receptors are accessible (MacGregor et al., 2019), as does inhibition of transferrin binding by Fab fragments (Steverding et al., 1995 [\[7\]](#)). It is likely that a combination of processes results in resistance to receptor antibody-mediated killing: first antibodies have to recognise parts of the receptor that are accessible in the VSG coat, this will be affected by the structure of the receptor and how much is occluded by ligand binding, a function of ligand concentrations, host species and receptor KD. Enough antibody has to be present on the cell surface for long enough to elicit a response. Two factors affect this, first the copy number of the receptor, ISGs are present at 50,000 to 70,000 per cell (Ziegelbauer et al., 1992 [\[10\]](#)) (<1% of VSG), TfR at a few thousand, but variable (Steverding et al., 1995 [\[7\]](#)), and HpHbR at 300 to 400 (Vanhollebeke et al., 2008 [\[11\]](#)). Second, the half time of the receptor antibody complex on the cell body plasma membrane will be short (Steverding et al., 1995 [\[7\]](#)) as the hydrodynamic forces rapidly push VSG-antibody complexes toward the flagellar pocket and endocytosis (Engstler et al., 2007 [\[12\]](#)). In summary, we provide evidence that GPI-anchored receptors in trypanosomes are not sequestered in the flagellar pocket but are distributed across the cell body and flagellum surface plasma membrane and this leads to re-evaluation of how trypanosomes avoid receptor antibody mediated clearance.

Materials and methods

Cell lines

T. brucei Lister427 bloodstream form cell lines were used throughout expressing either VSG221 (MiTat1.2) from BES1, VSG118 (MITat 1.5) from BES3 or VSG224 (MITat 1.3) from BES7. BES7 expressing cells were as described in Hertz-Fowler *et al.* (Hertz-Fowler et al., 2008 [\[13\]](#)) where BES1 and BES7 were tagged with Puromycin and G418 resistance genes respectively. Cells were maintained in culture in HMI-11 medium (Hirumi & Hirumi, 1989 [\[14\]](#)). For ligand-uptake assays and immunofluorescence cells were harvested in mid-log phase (0.5-1x10⁶ cells/ml). To allow further

modification by (CRISPR)-associated gene 9 (Cas9) gene editing, BES1 and BES7 expressing cell lines were first modified by transfection with a construct containing Cas9 and T7 RNA polymerase transgenes (Beneke et al., 2017 [DOI](#)).

Antibiotic selection was used when appropriate. The primers used to generate the cell lines are listed in Table S5 [DOI](#).

Constructs for ESAG7 +/- GPI attachment site

Constructs are represented schematically in Figure 2A [DOI](#) and described in brief below: BES7 constructs:

The presence of a G418 resistance gene upstream of ESAG7 was utilised to generate the required constructs from BES7 genomic DNA (Hertz-Fowler et al., 2008 [DOI](#)). A PacI-EcoRI PCR fragment from the promoter to the G418 resistance gene start was annealed to an EcoRI-PacI PCR fragment from the G418 stop codon through to the ESAG7 3' UTR. The resulting EcoRI site was used to insert a blasticidin resistance open reading frame. The entire construct was cloned into pGEM-T, allowing the required cassette to be excised by PacI for transfection into BES7 cells expressing Cas9, alongside suitable sgRNAs to target the BES7-ESAG7 region, designed by ChopChop (<https://chopchop.cbu.uib.no/> [DOI](#)). This construct contained the endogenous BES7 ESAG7 including its GPI anchor. To remove the GPI anchor addition sequences and express an ESAG7 similar to those in the other Lister427 bloodstream expression sites, a SacI-BsrBI fragment was excised and replaced with the corresponding fragment from BES1 ESAG7. This fragment was generated by RT-PCR from RNA isolated from BES1 VSG221 cells using standard methods.

BES1 constructs:

A similar approach was used to make ESAG7 modifying constructs from BES1 exploiting a puromycin resistance gene located between the promoter and the ESAG7 gene in the BES1 expression site (Hertz-Fowler et al., 2008 [DOI](#)). PacI-BstEII and BstEII-PacI PCR fragments were generated from genomic DNA, ligated and cloned into pGEM-T. To generate BES1-ESAG7 with a GPI anchor from the BES1 expression site, the distal SacI-PacI fragment was replaced by a PCR fragment encoding SacI-ESAG7 stop codon (blunt end) from BES7-ESAG7 DNA and a BES1-ESAG7 3' UTR region from the BES1 site corresponding to the BES7-ESAG7 stop codon (blunt end) to PacI. Constructs were linearised with PacI and transfected with sgRNAs targeting the ESAG7 gene in either BES7 or BES1 and electroporated into BES1 VSG221 cells expressing Cas9. BES7 modified cell lines were selected with blasticidin and BES1 modified cell lines with puromycin.

Changing the VSG from 224 to 221:

BES7-VSG224 expression was switched to VSG221 by transfection of a linearised plasmid fragment encoding VSG221, flanked by 5' and 3' VSG221 intergenic regions and a downstream G418 resistance gene flanked by alpha-beta tubulin intergenic regions.

Sequences targeting the 5' and 3' UTRs of the endogenous VSG224 gene were generated by PCR from BES7 (TAR153) DNA (a kind gift of Luísa Figueiredo) and cloned via KpnI/XhoI and SpeI/SacI fragments upstream and downstream of the VSG221-blasticidin expression cassette. G418 resistant clones were screened for binding to anti-VSG221 antibody through immunofluorescence assays and confirmed by Western blot analysis.

Protein gel electrophoresis and Western blotting

Protein samples were prepared and analysed by SDS-PAGE and western blotting following standard methods. Briefly, cells were harvested by centrifugation, washed in serum-free HMI-9 and resuspended in Protein Sample Buffer (10% glycerol, 2% w/v SDS, 80 mM Tris-HCl pH 6.8), at a concentration of 2×10^8 cells/ml. Immobilon-P membrane (Millipore) was used to retain the proteins. The primary antibodies used during western blotting are rabbit anti-FHR (Macleod et al., 2020 [DOI](#)) and rabbit anti-TfR (a kind gift of Piet Borst). Polyclonal anti-VSG221 and anti-SCD6 antibodies were generated from recombinant *E. coli* expressed proteins, injected into rabbits by

standard methods (Covalab). Secondary antibodies used were Peroxidase-conjugated AffiniPure Goat Anti-Rabbit IgG (H+L) (Jackson ImmunoResearch) and Alexa Fluor 488-Donkey Anti-Rabbit IgG (H+L) (Invitrogen).

Membranes were blocked in Tris-buffered saline (TBS), 0.1% Tween20, 2% w/v Marvel dried milk and probed in appropriate dilutions of primary and secondary antibodies. Washing steps were performed with TBS, 0.1% Tween20 and blots were analysed on a BioRad ChemiDoc system, ensuring signals remained below saturation limits. Quantitation of transferrin receptor protein level was performed using Fiji-ImageJ software (<https://imagej.net/software/fiji/>), with background subtraction and comparison with intensity measurements taken from a protein dilution series from BES7 expressing cells.

RNAseq

mRNA was sequenced using Illumina paired-end sequencing technology by the Beijing Genomics Institute. Raw reads were deposited to ENA under the accession number PRJEB90063. Analysis was performed using Galaxy (www.usegalaxy.org) (Afgan et al., 2018). Reads were quality filtered using Trimmomatic (Bolger et al., 2014). Sequences were searched for all common Illumina adaptors and the settings used were: LEADING:10 TRAILING:10 SLIDINGWINDOW:5:15 MINLEN:50. mRNA abundance was calculated using SalmonQuant with default settings and *Trypanosoma brucei* Lister strain 427 2018 v68 reference transcriptome and genomes from TriTrypDB (Aslett et al., 2010).

Ligand uptake

Bovine holo-transferrin (ThermoFisher) was dialyzed against PBS and labelled with Alexa fluor 568 NHS ester (ThermoFisher) following the manufacturer's protocol. Canine transferrin was obtained from Abbexa and loaded with iron using a four-fold molar excess of ammonium iron (III) sulphate in presence of 5 mM sodium bicarbonate followed by dialysis against 10 mM HEPES pH 7.4, 150 mM NaCl as described previously (Trevor et al., 2019)(Trevor et al., 2019). Canine holo-transferrin was then labelled with Alexa fluor 568 NHS ester (ThermoFisher) as above. The Hp-Hb complex was purified as described (Lane-Serff et al., 2014) and labelled with Alexa fluor 568 NHS ester (ThermoFisher) following the manufacturer's protocol.

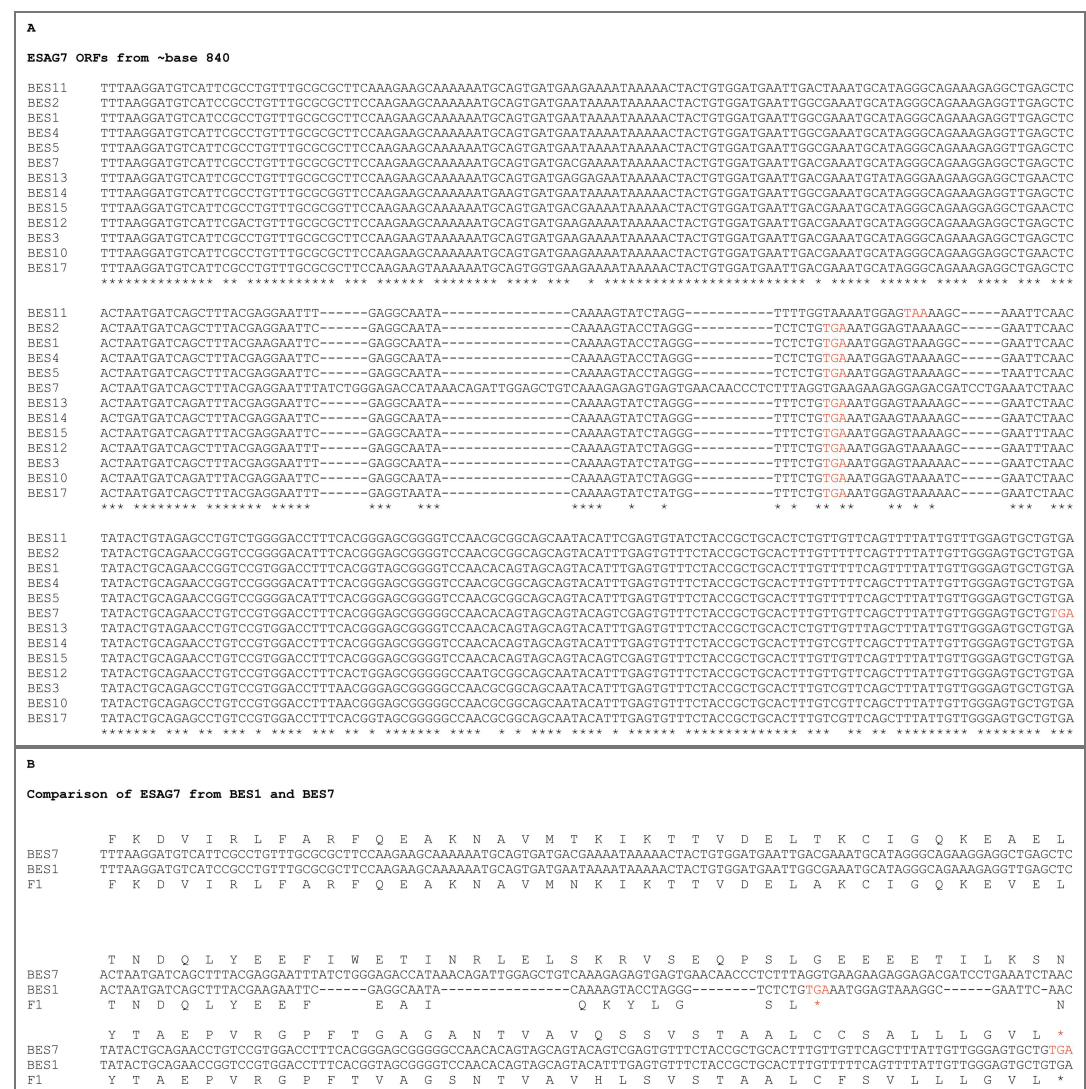
Mid log-phase cells ($\sim 6 \times 10^5$ /ml) were supplemented with 500 nM Alexa 568 labelled ligand and incubated at 37°C. At time points, cells were fixed by the addition of 8% formaldehyde in PBS to a final concentration 1% and incubated for 5 min. Cells were further washed with PBS plus 1% BSA and DNA was stained with Hoechst 33258. Slides were prepared and images were captured within next within the next 30 min using Zeiss Axio Imager.Z2 with X100 objective. Samples in any one experiment were excited with the same intensity and using the same exposure time. Images were processed using the ZEN Blue software.

Immunofluorescence localisation and microscopy

T. brucei cells were fixed in culture by the addition of one volume of 8% formaldehyde, 0.4% glutaraldehyde (EM grade TAAB Laboratories Equipment Ltd.) in PBS for 10 min. After fixation, cells were washed with serum free HMI-9 plus 1% bovine serum albumin (BSA) twice and immobilized on poly-lysine coated slides for 1 hr. Immobilized cells were treated with 0.1% triton X-100 (in PBS) for 5 min, washed with PBS and then incubated with PBS, 1% BSA for 10 min. Cells were incubated with the rabbit anti-TfR (1:50) or anti- FHR (Macleod et al., 2020) (1:50) in PBS, 1% BSA for 1 hr, washed three times with PBS and then with donkey anti- rabbit Alexa488 (1:800) in PBS, 1% BSA for 1 hr. Cells were washed three times with PBS and the last wash contained 0.1 µg/ml Hoechst 33258 to stain DNA. Cells were mounted with anti-fade and were visualized under Zeiss Axio Imager.Z2 with X100 objective. In any one experiment, samples were illuminated with same exposure time, intensity for respective excitation channels. All immunofluorescence experiments were repeated three to six times with similar results. The images were processed with ZEN Blue software by adjusting the brightness and contrast with the same changes being made to all images in any one experiment including the negative controls. Individual cells were

manually outlined to define the whole-cell regions of interest (ROI), and background fluorescence was measured from cell-free regions within the same field using the ZEN Blue software and it was subtracted. For each condition, mean fluorescence intensity (arbitrary unit) was quantified from at least 50-100 cells across the repeated experiments. Mean fluorescence intensity was measured and background fluorescence intensity was subtracted. ImageJ software was used to measure the point fluorescence intensity of ligands and receptor antibodies by drawing a line perpendicular to the long axis of the cell, and the distribution profile against the distance was plotted. For 3D structured illumination, a Zeiss Elyra7 microscope with X63 objective was used in the Cambridge Advanced Imaging Centre. Slides were prepared as above and the processing of all images and the 3D reconstruction were performed with ZEN Blue software only by adjusting the brightness contrast. All raw and processed images are available at EMBL-EBI BioImage Archive accession S-BIAD2480.

Supplementary figures

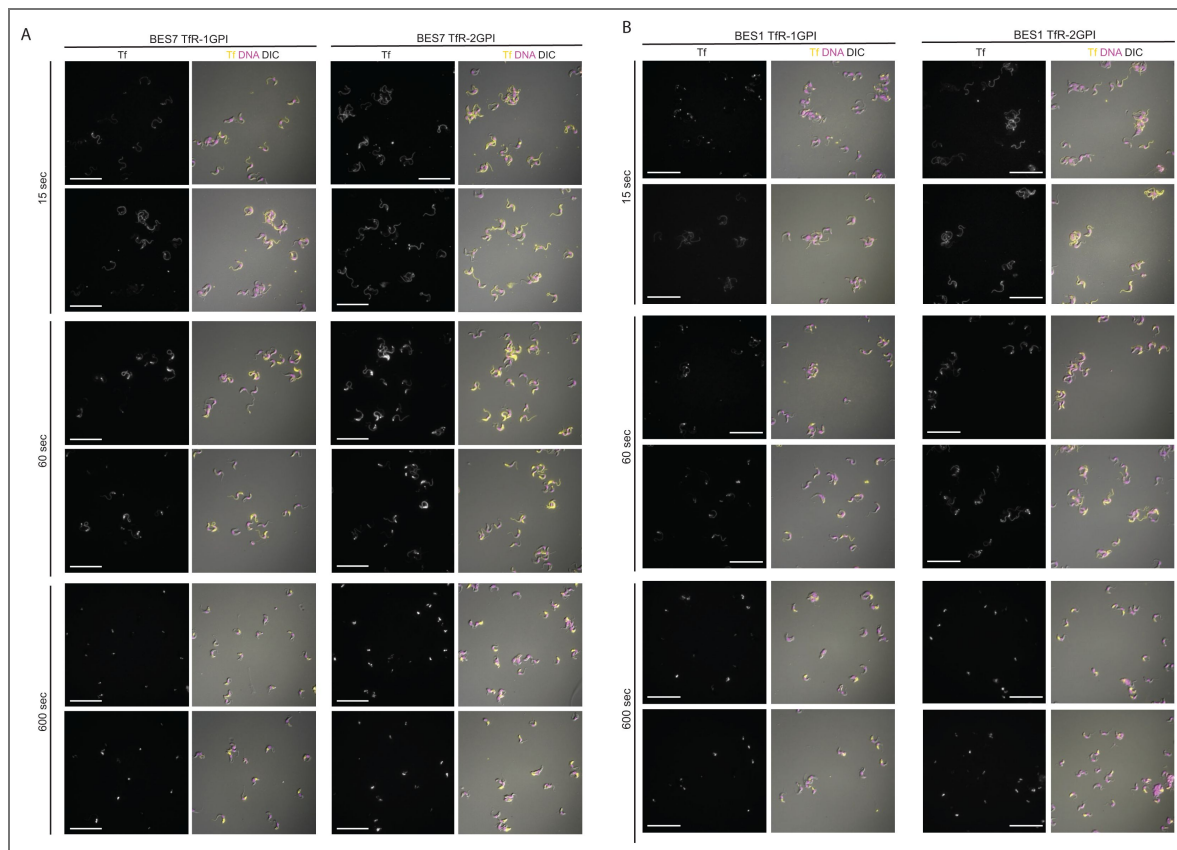


Supp Figure 1. Relationship between ESAG7 variants. A. Clustal W alignment of the nucleotide sequences of ESAG7s showing the translation termination codon in red. The sequence for GPI-addition is present in all genes but a series of small deletions between nucleotides 980 and 1080 resulted in a frame shift and a new stop codon that effectively removed the C-terminal ~50 residues from the open reading frame. B. Same alignment but just

including ESAG7 from BES1 and BES7 showing the conservation of the coding potential to the 3' side of the stop codon in BES1 ESAG7 indicating the shorter ESAG7 genes probably arose from a longer BES7-like variant. Functional stop codons are in red.

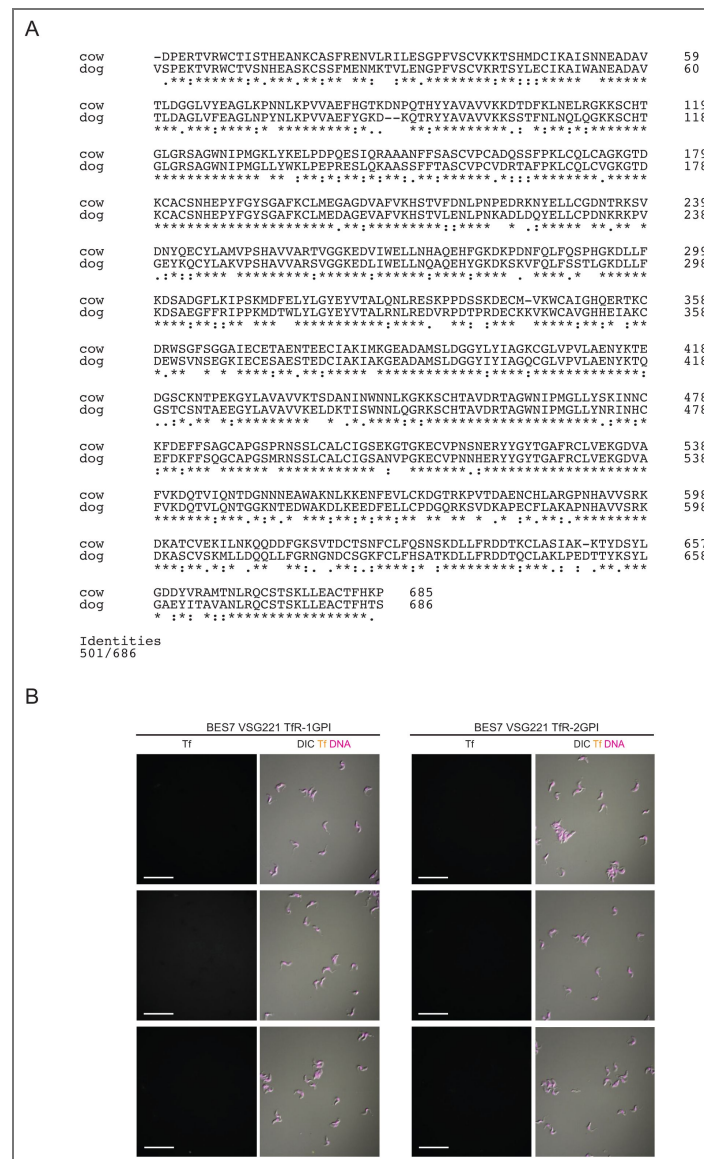
Supp Figure 2. Relationship between ESAG 6 and 7.

Banerjee, Minshall et al., 2025 *eLife* **14**:RP107191. <https://doi.org/10.7554/eLife.107191.2>



Supp Figure 3. Transferrin endocytosis is rapid, in both single and double GPI- anchored cells.

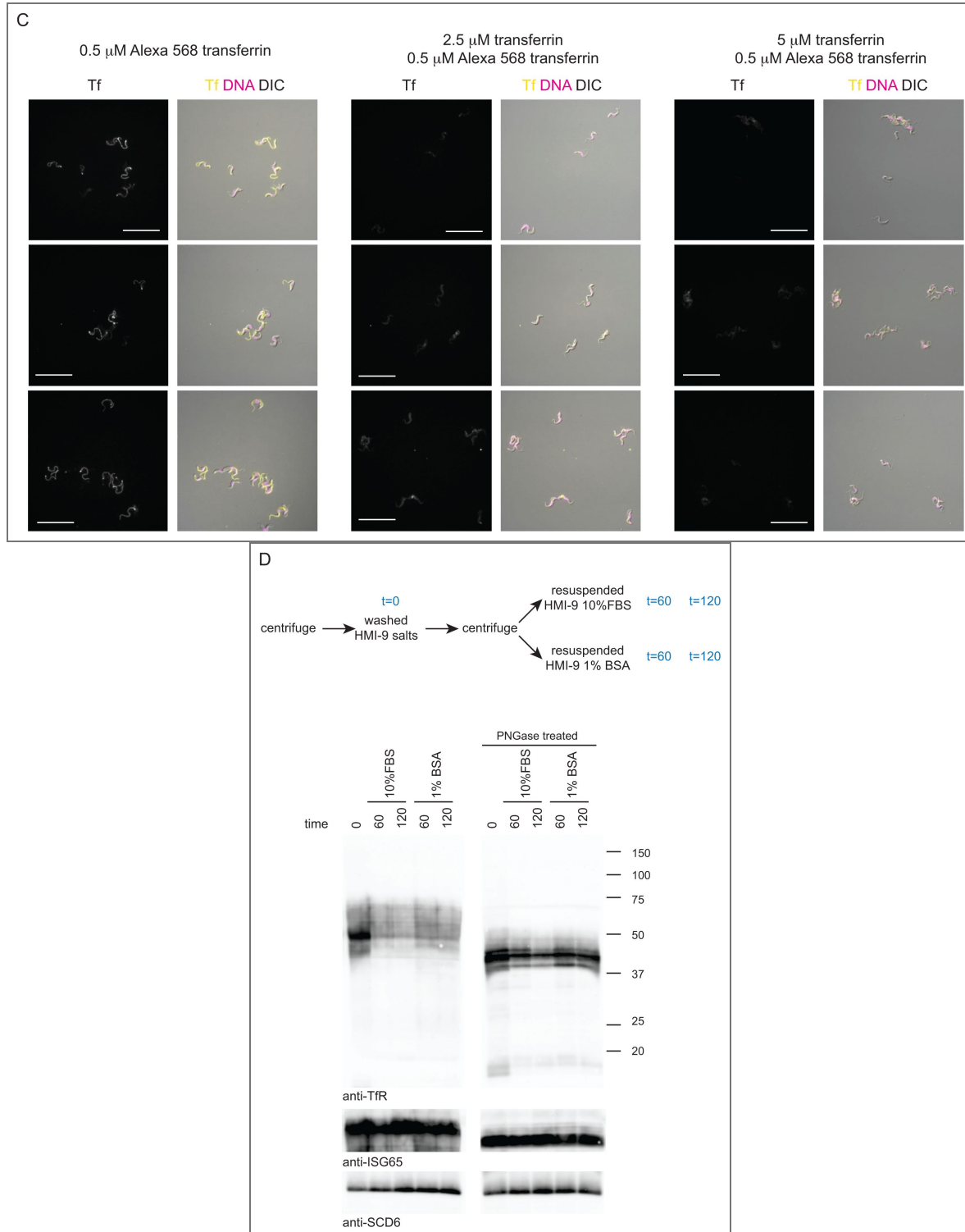
Initial binding is to the cell surface and internalization is detected within 60 sec. Larger fields of cells from the experiments shown in Figure 3. Scale bar is 20 μm.

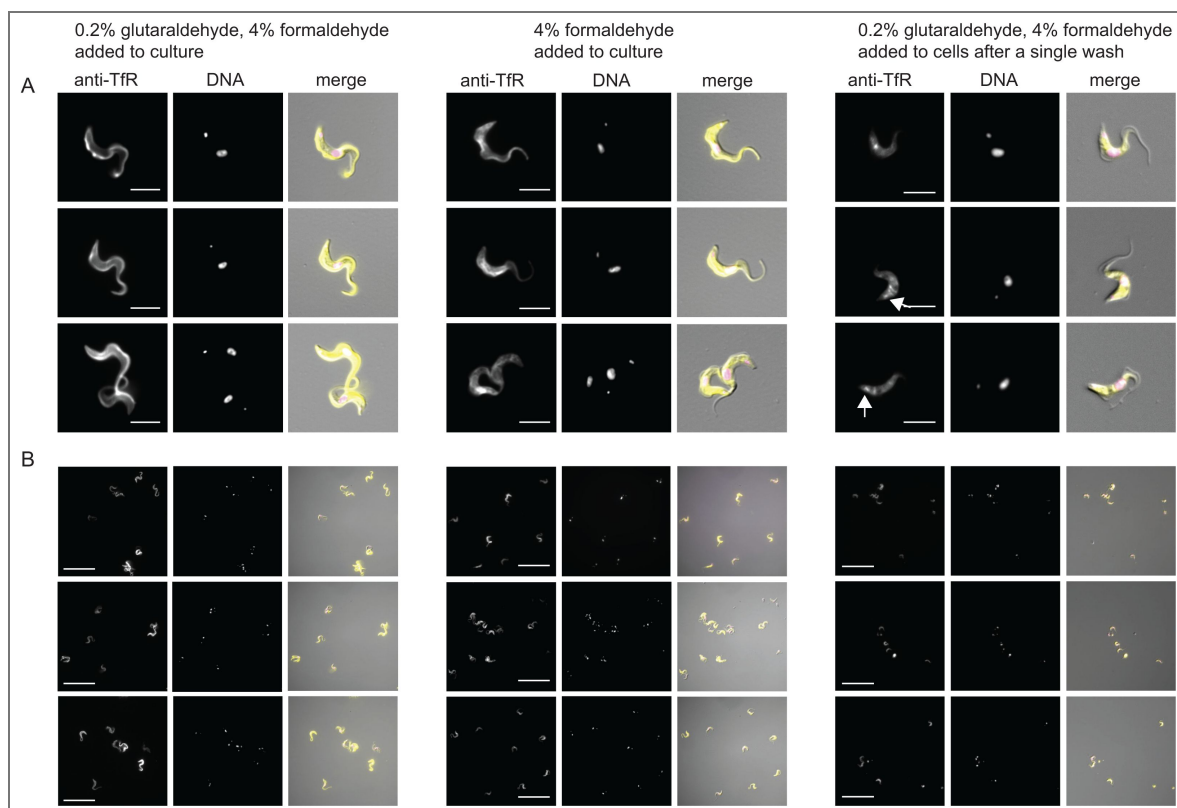


Supp Figure 4. Controls for specificity of bovine transferrin binding to *T. brucei* BE7 VSG221 transferrin receptors.

(A) Clustal W alignment of the polypeptide sequences of cow and dog transferrin. (B) Cultures were supplemented with 500 nM Alexa568 labelled canine holo- transferrin and incubated for 60 sec and fixed in culture by addition of formaldehyde to 1%. Following washing with PBS containing 1% bovine serum albumin cells were visualized under Zeiss Axio Imager.Z2 with x100 objective. Images for both BE7 VSG221 1- and 2-GPI Tfr cells were captured with same intensity of excitation and exposure time. Images showing the absence of any bound or endocytosed canine transferrin in around forty cells in population of both 1- and 2GPI Tfr cells. Scale bar is 20 μ m. Images were processed ZEN Blue 3.4 software. (C) Cultures of single (1-GPI) GPI-anchored Tfr expressing BE7-VSG221 cells were supplemented with either 0.5 μ M TfA568 only, 2.5 μ M unlabelled Tf plus 0.5 μ M TfA568, or 5 μ M unlabelled Tf plus 0.5 μ M TfA568, and incubated for 15 sec, then fixed by addition of formaldehyde for 5 min. DNA was stained with Hoechst. Following washing with PBS containing 1% bovine serum albumin cells were visualized under Zeiss Axio Imager.Z2 with X100 objective. Larger field views in all these experimental sets were captured with same intensity of excitation and exposure time. (D) BE7 cells in late logarithmic phase growth were counted by haemocytometer, harvested by centrifugation and treated as follows: Left panel: Lane 1 (t=0 FBS): washed briefly in serum-free HMI-9 and resuspended at 1×10^8 cells/ml in protein sample buffer; Lanes 2-5, washed briefly in serum-free HMI-9 and resuspended at 1×10^6 cells/ml in HMI-9 + 10% FBS (Lanes 2 and 3) or HMI-9 + 1% BSA (lanes 4 and 5) for a further 1 or 2 h prior to re-harvesting as described for lane 1. Right hand panel contains half the samples used for the left side but treated with PNGaseF prior to resuspension in protein sample buffer. 2×10^6 cells/lane were separated by SDS-PAGE on 15% gels. Western blot analysis was performed with anti-Tfr, anti-ISG65 and anti-SDC6 antibodies.

Supp Figure 4. (continued)

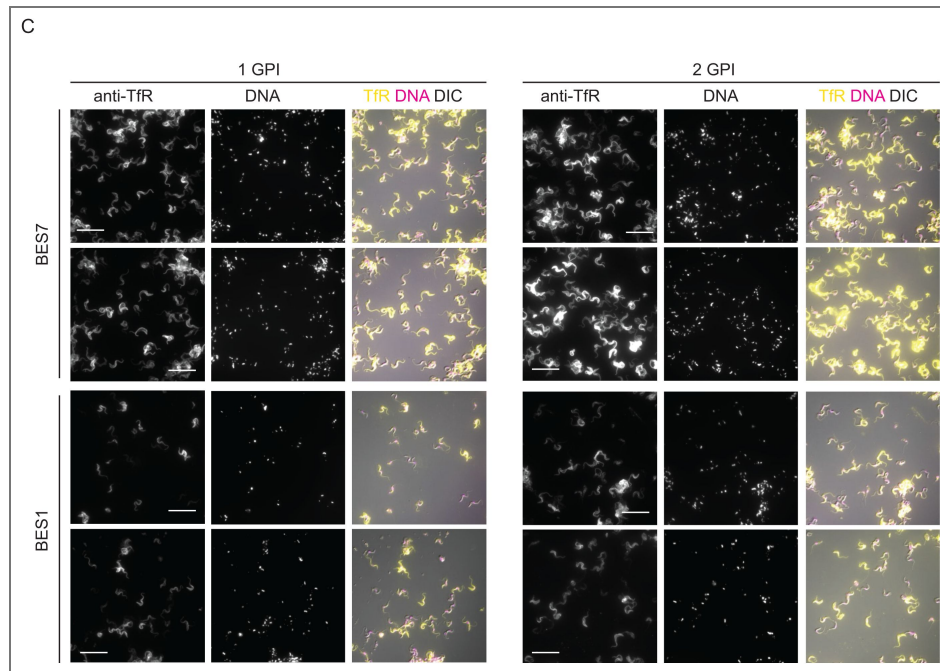




Supp Figure 5. Immunofluorescence localisation of single and double GPI-anchored transferrin receptors to the entire plasma membrane.

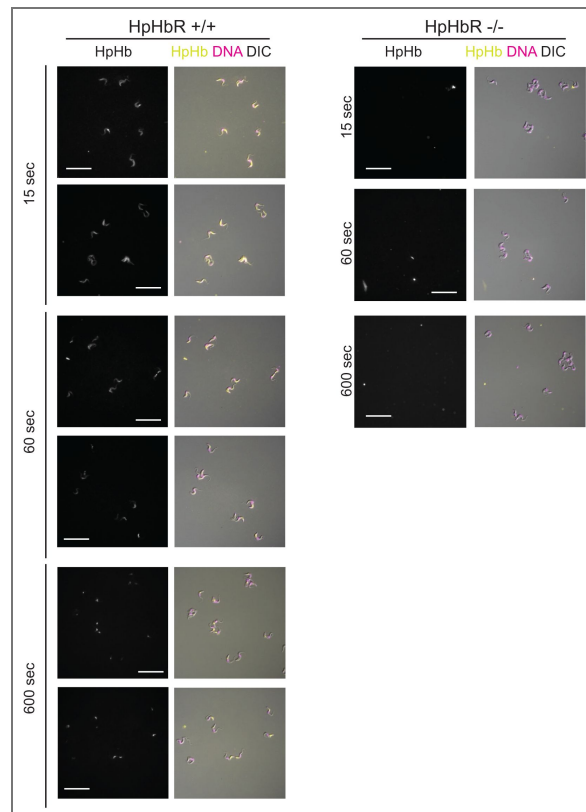
(A) Different fixation protocols for the localisation of single GPI-anchored TfR expressed from BES7. Cells were fixed in culture with final concentrations of either 4% formaldehyde plus 0.2% glutaraldehyde, 4% formaldehyde, or washed with (1X PBS+ 1% BSA) then fixed with 4% formaldehyde plus 0.2% glutaraldehyde. Immunofluorescence localisations were performed with rabbit anti-TfR followed by donkey anti-rabbit Alexa 488 and DNA was stained with Hoechst. A Zeiss Axio Imager.Z2 with X100 objective was used to visualise the cells. Images in all these experimental sets were captured with same intensity of excitation and exposure time. Scale bar is 5 μ m. Images were processed with ZEN Blue 3.4 software. (B) A larger number of cells is shown for Supp. Figure 5A. Scale bar is 20 μ m. (C) Larger fields of cells from the experiments shown in Figure 4. Scale bar is 20 μ m.

Supp Figure 5. (continued)



Supp Figure 6. Haptoglobin-haemoglobin endocytosis assay.

Initial binding is to the cell surface followed by rapid endocytosis. Larger fields of cells from the experiments shown in Figure 6. Scale bar is 20 μ m.



Supp Figure 7. The complement factor H receptor localises to the cell surface.

Larger fields of cells from the experiments shown in Figure 7B. Scale bar is 20 μ m.

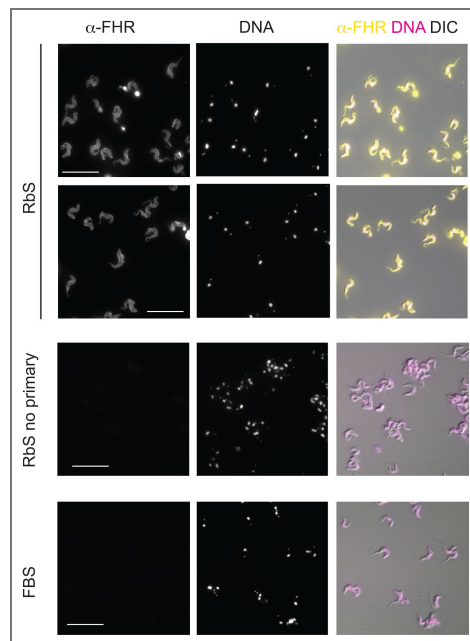


Table S1. VSG and Tfr mRNA abundances in transcripts per million transcripts for BES3, BES1 and BES7 cell lines

Gene	BES3 TPM	BES1 TPM	BES7 TPM	Geneid
BES3 VSG	76528	13	0	Tb427_000487400
BES1 VSG	5	71558	9	Tb427_000016000
BES7 VSG	5	1	72659	Tb427_000025600
BES1 ESAG6	3	567	8	Tb427_000014400
BES1 ESAG7	4	893	15	Tb427_000014300
BES2 ESAG6	3	189	3	Tb427_000016400
BES2 ESAG7.1	3	149	4	Tb427_000016200
BES2 ESAG7.2	4	130	5	Tb427_000016300
BES3 ESAG6	420	9	14	Tb427_000019200
BES3 ESAG7	897	17	39	Tb427_000019100
BES4 ESAG6	5	127	4	Tb427_000020600
BES4 ESAG7.1	6	72	6	Tb427_000020400
BES4 ESAG7.2	6	70	6	Tb427_000020500
BES5 ESAG6	11	140	8	Tb427_000022500
BES5 ESAG7	8	50	7	Tb427_000022400
BES7 ESAG6.1	6	4	1129	Tb427_000024300
BES7 ESAG6.2	4	6	1568	Tb427_000024400
BES7 ESAG7	5	6	2076	Tb427_000024200
BES10 ESAG6	51	4	11	Tb427_000005600
BES10 ESAG7	44	8	31	Tb427_000005500
BES11 ESAG6	4	3	29	Tb427_000006100
BES11 ESAG7	5	4	10	Tb427_000006000
BES12 ESAG6	11	17	29	Tb427_000007400
BES12 ESAG7	16	12	41	Tb427_000007500
BES13 ESAG6	7	7	24	Tb427_000008500
BES13 ESAG7	7	6	16	Tb427_000008600
BES14 ESAG6	5	19	24	Tb427_000009700
BES14 ESAG7	10	20	62	Tb427_000009600
BES15 ESAG6	15	41	146	Tb427_000011200
BES15 ESAG7	48	89	198	Tb427_000011100
BES17 ESAG6	78	9	26	Tb427_000013100
BES17 ESAG7	32	15	60	Tb427_000013000
ESAG6	622	1142	3024	
ESAG7	1096	1542	2573	
ESAG 6+7 total	1718	2683	5597	
ESAG 6+7 total/VSG	0.022	0.038	0.077	

BES7 1GPI

15sec			60sec			600sec		
Tf signal	Background	Corrected	Tf signal	Background	Corrected	Tf signal	Background	Corrected
270.527013	199.470774	71.0562392	307.898051	255.83777	52.0602813	298.18776	247.517913	50.669847
216.021364	191.24691	24.7744536	534.819246	273.520626	261.29862	421.620797	259.611457	162.009339
225.960869	187.769207	38.1916624	352.109405	271.572738	80.5366671	525.067093	275.33099	249.736102
223.6437	191.168308	32.4753919	422.856287	268.70876	154.146527	404.222876	264.124944	140.097932
240.177945	197.368701	42.8092434	526.585492	267.44094	259.144551	317.201097	265.283698	51.9173991
223.787773	196.606216	27.1815571	326.276981	268.906205	57.3707761	354.472614	257.252472	97.2201413
211.061524	195.937614	15.12391	336.227911	247.643988	88.5839226	281.166984	263.821168	17.3458156
285.609647	202.17503	83.4346177	562.718023	264.388187	298.329836	545.337859	265.13532	280.202539
319.135188	204.516947	114.618241	526.355311	279.857909	246.497402	279.489983	247.529167	31.9608164
291.465779	203.171943	88.293836	481.368734	277.767392	203.601343	329.179929	279.043807	50.1361218
227.405588	196.658576	30.747012	418.053302	264.294802	153.758499	368.099377	287.520689	80.5786888
273.039846	197.471913	75.567933	520.491879	281.689409	238.80247	418.60365	283.052773	135.550878
227.676158	198.695485	28.9806723	725.054756	292.028804	433.025953	296.116309	272.594014	23.5222947
310.921808	199.040633	111.881175	342.69834	269.357722	73.3406181	368.292317	296.640416	71.6519005
329.087392	199.745067	129.342325	722.31927	303.038324	419.280946	313.839668	287.690597	26.149071
325.985932	202.776398	123.209534	365.881844	251.106091	114.775753	348.65734	282.90712	65.7502206
303.677856	201.608826	102.06903	342.563364	243.340556	99.2228087	324.305968	286.770399	37.5355684
305.582946	193.846737	111.736209	297.669645	250.079745	47.5899006	389.365908	292.704202	96.6617058
277.370995	267.047632	10.323363	370.522424	253.168417	117.354007	314.635338	288.075947	26.5593912
329.228269	276.67244	52.5558294	539.232728	254.509029	284.7237	302.14597	277.028904	25.1170662
291.481791	281.910248	9.57154212	740.690224	270.793639	469.896585	368.235532	270.454079	97.7814532
309.974519	285.053016	24.9215029	385.107082	254.47228	130.634803	328.777435	275.477417	53.3000174
380.338358	288.081158	92.2572005	759.793115	288.559912	471.233203	445.845563	277.09124	168.754323
311.20289	284.48712	26.7157704	765.506649	267.375435	498.131214	534.611116	313.190433	221.420683
314.588837	283.180035	31.4088025	290.219871	244.393147	45.8267239	323.518369	302.143756	21.3746136
299.169318	285.910917	13.2584012	583.363392	290.711277	292.652115	571.593062	338.558758	233.034304
320.984716	278.742652	42.2420641	752.483352	257.612768	494.870584	379.215239	316.200939	63.0143
291.903978	273.07962	18.8243582	605.705147	240.238286	365.466861	411.69931	322.19736	89.5019505
311.865041	258.063849	53.8011916	333.264269	212.922133	120.342136	395.281697	309.562846	85.7188511
277.344201	274.748193	42.5960088	576.087851	233.959789	342.128063	365.18531	322.924761	44.2605489
306.910005	275.862761	31.0472436	305.344442	216.243535	89.1009066	484.290122	322.056405	162.233716
332.822617	274.593362	58.2292545	508.45768	230.659728	277.797952	369.286934	315.739937	53.5469972
316.434873	283.666085	32.7687881	249.85369	214.239243	35.6144462	366.540495	335.080743	31.4597517
301.136478	280.808377	20.3281001	248.20127	196.615315	51.5859552	682.353022	345.125028	337.227995
322.66434	278.591551	44.0727891	256.09504	203.378822	52.7162184	331.73847	329.167281	2.57118897
297.061457	264.237987	32.8234699	561.548287	209.296137	352.252149	421.569698	347.413535	74.1561633
323.003802	273.542228	49.461574	349.850106	221.123158	128.726948	571.021838	342.877755	228.144083
321.657315	277.6991	43.9582149	421.472867	222.62335	198.849517	463.555455	360.269772	103.285684
295.111735	284.313505	10.7982302	579.987291	232.891397	347.095894	434.58454	360.08541	74.4991301
295.079785	287.06964	8.01014523	273.186964	230.1948	42.9921639	549.715193	368.631464	181.083729
297.655363	286.472762	11.1826011	362.610993	243.257201	119.353793	394.304326	366.045671	28.258655
314.842956	279.825539	35.0174173	270.244238	210.204832	60.0394065	369.846281	343.532459	26.318218
283.503671	268.304394	15.1992765	333.347274	217.350429	115.996846	397.938942	360.525961	37.4129808
286.783833	264.197641	22.5861923	283.427531	234.508666	48.9188651	376.154727	355.099136	21.0555911
302.163543	269.87171	32.2918331	275.043757	233.051081	41.9926755	515.567764	370.928684	144.63908
306.631931	275.414341	31.2175892	705.877268	238.326194	467.551074	449.781633	346.985079	102.796553
340.009019	259.754924	80.2540956	350.963646	227.040899	123.922747			
271.998745	262.179116	9.8196288	298.887651	227.831518	71.0561323			
			563.720679	244.149112	319.571567			
			366.481103	229.013987	137.467116			
			580.596634	261.257964	319.33867			
			306.839855	258.587289	48.2525663			
			636.904073	252.959962	383.944111			
			452.088293	245.698578	206.389715			
			322.987124	232.208649	90.7784752			
			732.011736	251.845656	480.166081			
			307.210517	225.64119	81.5693262			

Table S2.

BES7 2GPI

15sec			60 sec			600 sec		
Tf signal	Background	Corrected	Tf signal	Background	Corrected	Tf signal	Background	Corrected
247.892737	182.852743	65.0399943	390.966746	225.475361	165.491385	606.63885	278.712675	327.926174
265.317564	184.88521	80.432354	467.652758	235.807767	231.84499	522.836779	278.919319	243.917461
237.91247	187.100238	50.8122317	739.189223	237.932359	501.256864	321.81492	268.368939	53.4459805
349.815799	188.069739	161.74606	565.400342	238.383329	327.017013	502.131997	273.638099	228.493898
293.908391	185.158769	108.749621	358.14173	224.732379	133.409351	500.773485	281.696054	219.077431
303.537896	188.214237	115.323658	287.355536	218.453149	68.9023869	357.373859	284.06719	73.3066693
242.402414	180.268968	62.1334459	630.247598	216.679795	413.567803	634.464334	293.587598	340.876736
282.917481	183.412698	99.5047824	430.27931	226.177157	204.102152	434.644566	288.377967	146.266599
244.126259	177.528166	66.598093	687.079315	247.523684	439.555631	358.443243	285.289154	73.1540891
318.052266	185.544796	132.507471	422.426521	229.018606	193.407915	705.008599	305.744685	399.263914
279.258024	177.741175	101.51685	323.106905	230.680488	92.4264168	407.802718	296.756594	111.046123
282.688793	180.297829	102.390965	817.271614	257.465417	559.806198	466.453458	284.082256	182.371202
307.713726	174.543191	133.170536	867.563407	245.309834	622.253574	411.330661	282.330703	128.999957
292.107308	180.116052	111.991256	784.141016	245.229815	538.9112	431.445288	265.008617	166.43667
234.642611	183.174794	51.4678169	915.109666	243.715578	671.394087	392.929535	244.140749	148.788786
256.771796	174.754545	82.0172507	576.869554	232.767271	344.102283	349.663426	255.51432	94.1491061
183.015015	168.348667	14.6663484	282.631245	210.943455	71.6877898	402.318674	241.763086	160.555588
277.032718	180.659628	96.3730898	279.903678	215.002284	64.901394	461.522451	268.549529	192.972921
257.322826	182.675541	74.6472853	387.094495	222.026336	165.06816	299.033233	261.283131	37.7501019
308.787712	187.524502	121.263211	339.57099	222.221949	117.349041	570.972593	269.795767	301.176826
188.163359	170.861471	17.3018883	656.628219	225.900496	430.727722	324.881755	258.292578	66.5891767
283.090492	199.312376	83.7781155	292.256795	223.84274	68.4140545	307.397299	262.124661	45.2726388
275.194135	198.538112	76.6560227	756.774718	261.302958	495.471759	322.427194	262.43369	59.9935046
333.272019	204.537505	128.734515	778.719406	232.517286	546.20212	451.68599	267.762544	183.923446
336.273948	202.756249	133.517699	598.660111	250.438023	348.222088	512.897845	249.290844	263.607001
300.131576	199.191786	100.939789	751.243739	258.717666	492.526073	509.51501	265.966233	243.548777
245.76248	200.218599	45.5438811	267.322421	226.966652	40.3557686	505.998385	270.275011	235.723375
273.39596	199.691615	73.704345	652.027004	246.496777	405.530227	563.237246	265.448661	297.788585
282.490435	196.436319	86.0541155	750.890727	249.877863	501.012864	550.74409	277.333799	273.410292
274.613809	199.458397	75.155412	450.811048	239.320912	211.490135	438.783627	261.282862	177.500765
338.689708	201.074637	137.61507	569.373968	249.068349	320.305619	555.228376	293.668578	261.559798
217.80695	196.792091	21.0148588	879.97008	243.141119	636.828961	721.591486	297.402217	424.18927
263.018576	199.732067	63.2865092	413.652013	278.43827	135.213743	387.920953	297.793329	90.1276244
249.496214	201.909242	47.5869727	322.634058	247.285841	75.3482165	369.137846	292.230747	76.9070994
316.877652	203.655118	113.222534	467.523137	249.116628	218.406509	503.786275	304.376169	199.410106
249.051887	202.983385	46.068502	819.823657	264.035084	555.788573	724.061185	304.643939	419.417245
281.69944	199.869596	81.8294843	678.35576	251.108023	427.247737	599.991469	315.178645	284.812824
266.194573	196.270174	69.9243994	322.547824	233.310124	89.2376992	416.201399	295.573689	120.62771
334.04686	203.385998	130.660861	801.099412	250.466419	550.632992	555.293517	309.033704	246.259813
299.349789	198.417214	100.932575	387.942382	242.532124	145.410258	596.003991	308.772722	287.23127
205.657151	191.041912	14.6152391	310.612123	215.097435	95.5146882	797.167341	311.801052	485.366289
289.521839	206.504662	83.0171762	721.486946	240.409111	481.077835	751.143348	303.746017	447.397331
227.34385	210.638597	16.705253	555.552117	235.715747	319.83637	552.304002	305.079011	247.22499
308.169623	208.847566	99.322057	703.015298	250.441654	452.573644	617.449004	292.164119	325.284885
262.07769	213.204104	48.8735866	361.444873	222.451855	138.993018	381.620313	279.732202	101.888111
267.468023	210.384198	57.0838249	329.944877	227.018772	102.926104	378.36234	290.088508	88.2738312
226.953046	211.141426	15.8116199	753.20834	232.552997	520.655344	437.47037	276.072788	161.397581
269.70887	207.039641	62.6692283	601.202009	244.113821	357.088188	435.313585	281.385836	153.927749
214.52326	204.283009	10.2402507	553.991433	233.478713	320.512719	340.448513	264.601207	75.8473066
243.931658	211.751512	32.1801461	276.938485	226.345573	50.5929123			
290.335054	213.014949	77.3201044	801.120032	240.220535	560.899497			
266.537327	201.642545	64.8947814	461.986787	220.874465	241.112321			
343.287173	205.822294	137.464879	563.446804	224.556551	338.890253			
216.982971	206.579469	10.4035027	693.986995	238.446832	455.540162			
249.042373	205.788136	43.2542373	624.868348	214.203814	410.664534			
229.234525	204.736721	24.4978039	312.007538	231.587357	80.4201808			
243.635383	205.122937	38.5124466	598.482561	212.61358	385.868981			

Table S2. (continued)

Table S3.

BES1-1 GPI								
15 sec			60 sec			600 sec		
Tf signal	Background	Corrected	Tf signal	Background	Corrected	Tf signal	Background	Corrected
234.775782	203.873377	30.9024048	430.576336	283.904626	146.67171	270.642614	242.393419	28.2491948
304.145806	202.660483	101.485323	427.849424	292.333524	135.515899	287.451875	240.639326	46.812549
214.218142	197.734326	16.483816	391.214386	298.413111	92.8012754	534.720136	257.636665	277.083471
291.23726	207.828409	83.4088513	358.875593	288.150767	70.7248261	277.984139	240.199136	37.7850026
261.148777	197.274474	63.8743035	414.639075	285.790569	128.848506	587.766636	250.113912	337.652724
220.735959	203.537313	17.1986453	414.424537	302.779975	111.644561	286.882546	254.604694	32.2778512
257.353716	193.027339	64.3263775	321.398208	292.160018	29.2381899	271.128827	249.529625	21.5992017
286.7046	247.012705	39.6918959	440.18949	299.598491	140.590999	277.848127	259.932282	17.9158458
271.928127	245.256637	26.6714894	356.546342	274.067394	82.4789482	388.457951	249.883058	138.574893
316.794736	258.311931	58.4828052	348.675102	282.865474	65.8096285	433.709883	253.86544	179.844443
268.456657	245.33976	23.1168969	345.528921	281.680444	63.848477	472.246897	239.098758	233.148139
297.862268	254.079599	43.7826682	394.034437	281.685222	112.349215	262.824643	249.630857	13.1937853
261.914497	244.681209	17.2332873	385.309148	278.705608	106.60354	520.444459	255.411644	265.032814
267.797693	241.958001	25.8396917	363.551895	281.644037	81.907858	547.705794	289.539683	258.166111
269.838301	255.565208	14.2730923	443.176529	263.135972	180.040557	537.976087	278.975919	259.000168
214.205674	246.193758	68.0119163	349.053596	273.279592	75.7740033	402.977283	291.006563	111.97072
249.678541	237.186589	12.4919523	421.862759	267.595096	154.267663	391.668459	300.948794	90.7196652
363.778742	247.94747	15.8312723	398.075086	226.932226	171.14286	397.115995	303.76716	93.3488352
301.326998	274.136265	27.1907331	351.621831	220.223631	131.3982	567.662969	291.537451	276.125518
272.657941	250.799173	21.8587675	470.560685	333.376129	237.184556	432.437619	292.397987	140.039632
271.73915	249.966658	21.7724924	302.103221	230.452781	71.6504402	619.407966	285.106685	334.301282
357.304406	334.772627	22.5317799	291.686809	222.739557	68.947252	402.32142	342.111073	60.2103475
341.010476	323.146652	17.8638236	425.499256	223.205339	202.293917	453.868563	360.021989	93.8465745
338.43351	325.604066	12.8294439	506.685785	215.680521	291.005264	405.981197	350.033441	55.9477564
343.671106	332.412443	11.2586625						
352.478644	322.30651	30.1721341						
BES1-2 GPI								
15sec			60 sec			600 sec		
Tf signal	Background	Corrected	Tf signal	Background	Corrected	Tf signal	Background	Corrected
218.399832	191.124987	27.2748454	724.43953	217.917581	506.521949			
294.154852	196.544623	97.6102287	548.99508	212.350592	336.644488	541.016991	262.699051	278.31794
212.136693	195.214351	16.9223422	314.431469	204.743319	109.68815	285.584734	260.479405	25.1053299
262.848476	194.241362	68.6071136	492.191051	210.504542	281.686509	523.488474	264.453981	259.034492
299.164106	194.355854	104.808253	445.269671	209.295051	235.97462	418.35013	270.350725	147.999405
248.558701	197.284171	51.2745293	447.533955	207.612561	239.921394	515.089333	249.04283	266.046503
335.442985	195.705871	139.737113	748.321073	206.81923	541.501843	698.288118	275.387093	422.901025
223.67045	192.648126	31.0223234	737.923592	205.295989	532.627603	373.834541	274.222351	99.6121898
213.017357	188.502244	24.515113	618.600222	207.961365	410.638857	297.319077	274.597894	22.7211826
245.347943	182.301101	63.0468418	557.782201	198.055422	359.726779	302.421563	278.956335	23.4652277
222.28335	187.05765	35.2257004	475.835383	207.339708	268.495674	392.204996	275.514891	116.690105
262.664858	190.868765	71.7960929	675.614144	210.883379	464.730765	298.669704	274.819244	23.8504597
222.551139	191.517776	31.0336628	370.293466	209.769431	160.524036	281.714589	266.892605	14.821984
240.715427	187.335455	53.3799722	437.232891	193.668192	243.564699	621.809336	273.515719	348.293617
264.645648	186.379458	78.26619	257.117092	196.785309	60.331783	286.954908	273.528366	13.4265421
279.265136	190.411315	88.8538216	643.597643	243.528223	400.06942	433.496754	266.8967	166.600054
268.783581	194.517592	74.2659891	527.445209	240.96208	286.483128	420.19317	273.223489	146.96968
242.429856	198.959583	43.4702732	467.698995	229.328577	238.370418	290.177674	262.666819	27.5108543
282.006158	194.093792	87.9123662	263.909656	217.802492	46.107164	341.553629	263.003851	78.5497775
249.620311	199.163133	50.4571789	330.397118	239.601499	90.7956182	426.453288	277.979573	148.473715
236.263425	197.902823	38.3606027	748.09636	239.377204	508.719156	466.51363	275.520429	190.993201
279.750019	193.188704	86.5613153	436.886298	249.448428	187.43787	491.687534	264.245959	227.441575
292.408658	195.500198	96.9084602	273.319074	258.868209	14.450865	429.519163	267.60496	161.914202
			376.5206	257.25119	119.269411	328.923152	273.940557	54.9825955
			720.093001	230.73258	489.360421	320.968958	266.208676	54.7602827
			795.160401	241.603944	553.556457	296.599195	279.496228	17.1029671
			378.107244	221.609327	156.497917	445.503247	280.268316	165.234931
			348.532029	237.285133	111.246958	469.347182	277.758576	191.588606
			358.095546	236.25002	121.845526	296.463426	270.560027	25.9033986
			511.25555	222.830525	288.425026	514.763865	264.364088	250.399778
			261.156128	231.916574	29.2395536	509.202331	275.454572	233.747759
			759.169984	221.642662	537.527322	504.4988	270.451083	234.047718
			784.147973	216.540623	567.60735			
			277.338981	233.243562	44.0954191			

BES1-1GPI			BES1-2GPI		
FI of cell	FI background	Corrected	FI of cell	FI background	Corrected
4169.1891	1369.28354	2799.90556	7397.90542	1601.33761	5796.56781
5517.00505	1518.63807	3998.36698	5194.63503	2016.5935	3178.04153
4466.68632	1366.80352	3099.8828	5046.05	1414.73603	3631.31397
4454.89058	1355.07642	3099.81416	7014.50272	1602.20093	5412.3018
3703.71709	1403.40315	2300.31394	9167.4573	1671.26481	7496.19249
3883.55348	1423.43266	2460.12082	4705.50447	1980.2286	2725.27587
4161.70658	1313.01821	2848.68837	5314.19549	1671.40467	3642.79082
5345.77071	1237.77552	4107.99519	5140.15445	1549.04497	3591.10948
3812.64022	1395.97521	2416.66501	6865.54536	1434.11577	5431.42959
5233.52321	1460.97417	3772.54904	5011.45679	1530.49063	3480.96617
4416.26956	1305.2035	3111.06606	5880.58125	1596.30831	4284.27294
5690.69117	1456.09166	4234.59951	5714.74356	1723.14111	3991.60244
4974.73575	1634.37369	3340.36206	4558.88611	1614.53589	2944.35022
4159.68304	1728.30496	2431.37908	6957.89341	1697.69679	5260.19663
5966.39615	1562.48411	4403.91204	4531.5057	1858.15333	2673.35237
6844.52143	1658.0438	5186.47763	4075.3613	1559.91582	2515.44548
4849.82735	1290.27353	3559.55381	6386.4478	1525.08275	4861.36505
4387.01769	1368.57291	3018.44478	4768.65125	1532.22332	3236.42793
4039.22212	1581.33536	2457.88675	5121.03961	1810.44301	3310.5966
5653.91974	1854.64047	3799.27927	4289.88786	1661.7653	2628.12256
6128.88791	1760.47869	4368.40921	4224.13117	1657.07444	2567.05673
5461.04714	2209.05985	3251.98729	4022.22367	1542.06811	2480.15556
4121.67164	2404.91355	1716.75809	4757.77671	1640.24423	3117.53248
4697.11634	1529.9975	3167.11885	3972.69131	1458.73595	2513.95536
4032.46619	2088.2814	1944.1848	4952.72435	1439.38026	3513.34408
4472.13665	1651.82691	2820.30974	4829.96955	1417.7771	3412.19246
4666.65855	1196.71066	3469.94789	4799.97044	1399.56123	3400.40921
6836.58984	1242.24821	5594.34163	6406.86227	1455.44769	4951.41458
4340.12546	1147.18195	3192.94351	5247.45768	1510.92269	3736.53499
4544.95411	1208.12037	3336.83374	4812.53749	1432.02822	3380.50926
4244.87746	1238.19354	3006.68392	3891.33291	1322.36933	2568.96358
3591.77771	1674.24542	1917.53168	4230.59221	1453.39171	2777.2005
4342.65985	1209.49804	3133.16182	5191.87406	1473.93191	3717.94215
4607.30645	1101.59142	3505.71503	4181.71502	1518.48565	2663.22938
4303.83696	1400.3393	2903.49766	5322.50447	1662.18291	3660.32157
4061.93873	1506.87737	2555.06135	5375.1074	1330.79175	4044.31565
4602.69081	1395.96891	3207.29391	5076.07008	1336.94676	3739.12332
4478.11811	1502.67812	2975.43999	5146.07223	1568.86249	3577.20975
4903.37558	1312.0718	3591.30378	3651.2423	1499.63945	2151.60285
3691.56141	1389.23716	2302.32425	4230.18298	1501.29572	2728.88725
3363.79062	1299.70141	2064.08921	4648.14714	1444.06839	3204.07874
3387.93578	1428.83978	1959.096	4921.54447	1618.38973	3303.15474
3595.42411	1487.81455	2107.60956	5262.78448	1663.90231	3598.88217
3073.78033	1375.52312	1698.25722	5094.47869	1515.9482	3578.5305
2946.87813	1334.96877	1611.90936	5069.50906	1799.36771	3270.14136
3583.73492	1531.62643	2052.1085	4622.10492	1412.16133	3209.94359
4565.25277	1370.67542	3194.57735	4646.93306	1466.65244	3180.28061
3289.89102	1503.71898	1786.17204	4191.70167	1302.97995	2888.72173
3901.68966	1475.30914	2426.38053	3250.59116	1610.00582	1640.58534
4489.0743	1677.21025	2811.86405	5114.45282	2225.04777	2889.40506
3344.62854	1442.39829	1902.23025	4169.27925	1557.3028	2611.97645
3226.46594	1737.63654	1488.82939	5792.44183	1510.41649	4282.02534
4014.50712	1757.63624	2256.87088	4092.54011	1577.27939	2515.26071
4816.73036	1683.81031	3132.92006	3919.93385	1439.58637	2480.34748
3354.05243	2063.54544	1290.50699	3929.92664	1387.49081	2542.43583
3515.58823	1466.18232	2049.40591	3469.22981	1284.5608	2184.66901
3846.69126	1294.55445	2552.13681	3484.57208	1504.05021	1980.52186
3776.52195	1309.98233	2466.53962	6399.72065	1424.07604	4975.6446
3469.76282	1414.831	2054.93182	5678.3475	1760.17238	3918.17512
4986.73253	1276.04999	3710.68253			
3775.15461	1180.49606	2594.65855			
4504.38413	1081.42983	3422.9543			
2792.16911	1034.30053	1757.86859			
3021.98636	1350.46996	1671.5164			
3434.43803	1301.40369	2133.03433			
3546.77908	1272.6147	2274.16437			
3986.38403	1438.23311	2548.15092			

Table S4.

BES7-1 GPI			BES7-2GPI		
Cell FI	Background F Corrected		Cell FI	Background F Corrected	
12837.0266	4681.34276	8155.68384	14596.8045	3396.05044	11200.7541
17856.946	5163.70588	12693.2401	12827.5242	3393.0242	9434.49997
14578.8019	6733.43986	7845.16204	12743.1915	3332.9112	9410.28027
8257.59879	4489.82684	3767.77196	12563.7286	3331.35159	9232.37704
21507.8573	6313.37498	15194.4824	11161.5916	3277.68613	7883.9055
9684.06919	6390.32598	3293.7432	10628.5005	3184.04184	7444.45867
11574.0715	2055.70618	9518.36528	10364.0054	3162.29286	7201.71257
14178.8166	3363.52103	10815.2956	9940.49315	3076.5333	6863.95985
9229.06276	3063.70515	6165.35761	9574.09572	3051.88258	6522.21314
11373.0827	2629.73056	8743.35219	9342.3204	3027.09073	6315.22967
9459.26121	3335.64905	6123.61216	9010.88678	3011.91133	5998.97545
16907.3104	5001.21501	11906.0954	8607.00698	2963.3325	5643.67447
12640.7288	3904.67238	8736.05641	7853.71566	2955.74043	4897.97523
9608.94197	4574.54556	5034.39641	7065.93033	2879.59719	4186.33315
21016.9562	4752.91193	16264.0443	6760.38553	2828.94613	3931.4394
18762.9114	5075.52375	13687.3876	6125.49891	2805.08647	3320.41244
10587.7768	4418.42652	6169.35031	5915.60049	2774.39108	3141.20941
6315.10407	3887.97902	2427.12505	13164.126	4607.8443	8556.28172
5872.17781	4196.63916	1675.53865	14804.4068	4590.81764	10213.5892
6740.69995	4107.35088	2633.34907	12044.1318	5208.19234	6835.93951
22877.5275	5306.45609	17571.0714	19551.6645	5003.8709	14547.7936
8086.627	5258.43577	2828.19123	15227.5612	5833.93699	9393.6242
12070.3399	4203.20336	7867.13658	14260.7814	4752.39096	9508.39049
11069.5206	3399.07264	7670.44792	19235.7898	4988.06845	14247.7214
8798.09693	3788.9649	5009.13203	9781.47423	4129.59609	5651.87814
8394.35496	3715.74066	4678.6143	13489.69	3658.64074	9831.04925
8378.80007	4547.47284	3831.32724	16104.6728	4310.65582	11794.0169
7918.69538	3390.73456	4527.96082	15238.7876	4766.47593	10472.3117
11517.9798	3080.5122	8437.46759	11790.272	3839.99295	7950.27903
10405.1445	3515.78948	6889.35503	15302.1429	3715.72763	11586.4152
9353.17874	2897.44106	6455.73768	7879.73013	3866.95582	4012.77431
15463.0206	3867.05777	11595.9629	10672.888	3674.82336	6998.06459
8263.89373	2437.17601	5826.71772	10890.1711	3531.50056	7358.67057
7236.70061	3213.57139	4023.12922	10747.2814	4092.10536	6655.17607
4703.99833	2573.29191	2130.70642	12120.564	3893.12971	8227.4343
14961.5196	3747.41527	11214.1043	10693.786	3313.44112	7380.34485
13139.3319	3888.15671	9251.1752	11334.7401	4036.58485	7298.15529
17608.9766	4108.78449	13500.1921	9852.47699	4077.37446	5775.10253
10541.4171	3529.86644	7011.5507	13146.5762	4806.09194	8340.4843
7598.97705	3195.20233	4403.77473	12910.7939	4896.15198	8014.6419
15210.038	3511.77355	11698.2644	9639.8082	3884.35938	5755.44881
6002.27436	1336.80393	4665.47044	14823.3155	5220.65122	9602.66426
7559.97018	1619.01031	5940.95988	12602.9559	4471.4112	8131.54473
11302.1653	1938.84383	9363.32151	10275.7252	4019.51278	6256.21245
11072.582	1652.44181	9420.14021	12437.123	3568.96638	8868.45659
4273.02183	1673.77179	2599.25004	11381.6733	4956.19825	6425.47509
10248.0598	1698.39438	8549.66537	12961.427	3769.22951	9192.19746
6818.36613	1502.03382	5316.33231	28652.0939	7790.18995	20861.904
7678.34762	1599.25187	6079.09574	39810.3447	7348.82936	32461.5154
13070.9653	2013.20888	11057.7564	11230.3035	5746.28909	5484.01436
5902.91196	1489.5861	4413.32586	34734.4885	8056.44612	26678.0424
6371.34498	1500.68632	4870.65866	10087.011	6371.82538	3715.18559
8271.58377	1379.03512	6892.54865	22591.4696	7408.2788	15183.1908
11719.5397	1918.39672	9801.14298	29298.3307	6245.77872	23052.552
11442.8635	1962.45823	9480.40524	17178.1426	5795.58891	11382.5537
3919.90251	1285.24344	2634.65907	30589.019	8105.09675	22483.9222
8493.24379	1627.55856	6865.68524	10660.4765	5307.62018	5352.85631
10624.4969	4166.08267	6458.41427	23269.3716	4672.84322	18596.5283
8933.71896	3393.76976	5539.9492	15321.0247	5381.674	9939.35071
12578.264	3328.29907	9249.96496	18432.5793	6361.81345	12070.7658
9401.38217	3636.60645	5764.77571	22047.3234	5155.23628	16892.0871
17873.1515	4066.34463	13806.8069	44621.7256	10497.3445	34124.3811
14654.5717	2931.92614	11722.6456	14672.1238	6648.39164	8023.7322
8199.72123	2538.99357	5660.72766	25233.7494	9511.99941	15721.7499
10690.3571	2804.71569	7885.64144	28621.3122	9511.99941	19109.3128
17867.1895	3092.72876	14774.4608	21952.2258	8968.63297	12983.5928
6417.39227	2788.89755	3628.49472	17127.1369	9357.45498	7769.6819
6908.05726	2958.14219	3949.91507	41312.7102	12880.4142	28432.296

Table S4. (continued)

Table S4. (continued)

20683.4922	3047.92127	17635.5709	21815.4615	9429.58456	12385.877
6095.30543	2798.56066	3296.74476	7354.70445	3872.3488	3482.35565
14839.6706	2914.04669	11925.6239	27227.0988	7005.97022	20221.1286
13068.3758	3050.76952	10007.6053	20877.738	8990.7165	11887.0215
6706.24789	3121.48318	3584.75471	26307.1804	8765.73706	17541.4433
16044.9009	4262.1351	11782.7558	32094.9278	7321.51356	24773.4142
15395.3617	6158.93497	6236.42676	13807.2987	6878.01313	6929.28559
13498.7991	3595.3978	9903.40131	37369.6264	9009.61228	28360.0142
9703.24426	3741.26092	5961.98334	31935.4964	9140.15698	22795.3394
19649.345	4109.21333	15540.1317	16396.7827	8824.2721	7572.51062
17432.3565	4652.47975	12779.8787	40337.0674	10206.0609	30131.0065
15859.6605	3661.14234	12198.5181	18377.9973	10058.169	8319.82832
17979.2955	5041.82697	12937.4885	37444.5905	9070.05294	28374.5376
16472.5519	3590.97316	12881.5787	22141.4275	7954.17203	14187.2555
14309.8958	4121.26972	10188.6261	22922.965	7011.73101	15911.234
16235.584	4150.48332	12085.1007	24585.2531	7588.65098	16996.6021
17900.6796	4531.93406	13368.7456	13685.735	7217.58968	6468.14533
14657.4608	3777.59456	10879.8662	36560.9886	9776.42725	26784.5613
15432.3774	4341.38168	11090.9957	18183.3487	11097.7722	7085.57648
20632.1338	3979.56919	16652.5646	20765.1325	9151.98895	11613.1335
17866.4617	4037.67044	13828.7912	9734.20179	7437.75866	2296.44313
17459.0386	4786.22063	12672.8179	10807.2598	8079.42412	2727.83565
9560.24487	3368.6286	6191.61627	20203.9132	8409.65566	11794.2575
4606.1829	2927.84182	1678.34108	16519.0085	7678.60639	8840.40211
4596.79703	3020.6873	1576.10974	18698.6301	6087.98564	12610.6445
6663.79374	2985.62929	3678.16445	13361.1002	4958.97028	8402.12994
7239.77317	3475.27868	3764.49449	9484.52546	4920.36035	4584.16511
6663.68638	3501.51167	5162.17672	22564.3954	6224.63287	16339.7625
			10647.0543	6146.32952	4500.72476
			26330.8566	8926.81754	17404.039
			33165.9144	9949.59197	23216.3224
			25891.9188	8903.45542	16988.4633
			10112.603	6528.53747	3584.0655
			36326.5213	8593.36712	27733.1542
			35099.0257	7446.47981	27652.5459

Table S5. Oligonucleotides

		Forward	Reverse	Template
BES7-ESAG7 + GPI anchor	PacI-EcoRI	TGATTAATTAGGCTCTTGGCGCT	GTGGAATTCATCTTTGTTCAAT CAT	BES7 gDNA
	EcoRI-PacI	ATCGCCTTCTTGAAGAAATCTT	CAATTAATTAAGCGCATCAAC TTGCACC	BES7 gDNA
Blast R	EcoRI-EcoRI	CCCGAATTCATAGTGCCTTTGTCTCAAGAAG	GCAGAAATTCATTAGCCCTCCCA	pPOT v7 Blast-Blast
BES7-ESAG7 no anchor	SacI-BsrBI	GTTGAGCTCACTAATGATCAGC	CCCCGCTCCCGTGAAAGGT	RT-PCR from Lister427 Mitat1.2 purified RNA
sgRNA templates	5' G418 resistance gene	GAAATTAATACGACTCACTATAGGGAATAG CCTCTCCACCCAGGTTTATAGAGCTAGAAA TAGC		
	ESAG7 3'UTR	GAAATTAATACGACTCACTATAGGAACGTG TGAAGCCTGAAATGTTTATAGAGCTAGAAA TAGC		
BES1-ESAG7 no anchor	PacI-BstEII	TGATTAATTAGGCTCTTGGCGCT	GATGTCGAGCCCGACGCGCT G	BES7 gDNA (contains G418 resistance gene in BES1 expression site)
	BstEII-PacI	CCACGCGCCACCGTCGACCC	CAATTAATTAAGCGCATCAAC TTGCACC	BES7 gDNA (contains G418 resistance gene in BES1 expression site)
BES1-ESAG7 + GPI anchor	SacI-BES7- ESAG7 stop	GGAGGCTGAGCTCACTAATGATCAGC	TACAGCACTCCCAACAATAA AGCTGAAC	BES7-ESAG7 + GPI-anchor clone
	BES7-ESAG7 stop-PacI	AGGAAGGATGCGACAGAAGCTGC	CGTTGGGAGCTCTCCCATATG GTC	BES1-ESAG7 no anchor clone
sgRNA templates	5' UTR	GAAATTAATACGACTCACTATAGGAGTCAT GCATGAACGACACAGTTTATAGAGCTAGAAA TAGC		
	ESAG7 3'UTR	GAAATTAATACGACTCACTATAGGAACGTG TGAAGCCTGAAATGTTTATAGAGCTAGAAA TAGC		
VSG221 in BES7 switch	KpnI-XhoI	GCTGGTACCCAAAGATTTGTTTGGAGGAG	CGCCTCGAGGCTTGTTTGTTC TGATCTTA	BES7(TAR153) DNA
	SpeI-SacI	CTTACTAGTATTTTCCCCCTCAATTTCC TT	AAAGAGCTCGTGAATCAAAAA AAAAAAATTAAATTC	BES7(TAR153) DNA

Data availability

RNA sequence data were deposited to ENA under the accession number PRJEB90063. All raw and processed microscopic images are available at EMBL-EBI BioImage Archive accession S-BIAD248.

Acknowledgements

We would like to thank: Luísa Figueiredo for the gift of the cell line expressing VSG224 from BES7 and a BES7 TAR clone; Piet Borst for the transferrin receptor antibodies; Jay Bangs for a lot of fun discussions (but no fish); and the Cambridge Advance Imaging Centre for Super Resolution Microscopy.

Additional information

Funding

MC and MH are Wellcome Investigators (MC, 217138/Z/19/Z; MH, 220797/Z/20/Z). Initial funding for SB was from the Newton-Bhabha Fund.

Funding

Funder	Grant reference number	Author
Newton Fund (NF)	Newton-Bhabha Fund	Sourav Banerjee
Wellcome Trust (WT)	https://doi.org/10.35802/217138	Mark Carrington
Wellcome Trust (WT)	https://doi.org/10.35802/220797	Matthew K Higgins

Author ORCID iDs

Sourav Banerjee: <https://orcid.org/0000-0002-0716-3668>

Matthew Higgins: <https://orcid.org/0000-0002-2870-1955>

Mark Carrington: <https://orcid.org/0000-0002-6435-7266>

References

1. Afgan E., Baker D., Batut B., van den Beek M, Bouvier D, Cech M, Chilton J, Clements D, Coraor N, Grüning BA, *et al.* (2018) The Galaxy platform for accessible, reproducible and collaborative biomedical analyses: 2018 update. *Nucleic Acids Research* **46**:W537-W544 <https://doi.org/10.1093/nar/gky379>
2. Aslett M., Aurrecochea C., Berriman M., Brestelli J., Brunk B. P., Carrington M., Depledge D. P., Fischer S., Gajria B., Gao X., *et al.* (2010) TriTrypDB: a functional genomic resource for the Trypanosomatidae. *Nucleic Acids Research* **38**:D457-62 <https://doi.org/10.1093/nar/gkp851>
3. Banerjee S., Minshall N., Webb H., Carrington M (2024) How are Trypanosoma brucei receptors protected from host antibody-mediated attack?. *BioEssays : News and Reviews in Molecular, Cellular and Developmental Biology* e2400053 <https://doi.org/10.1002/bies.202400053>
4. Bangs J. D., Andrews N. W., Hart G. W., Englund P. T (1986) Posttranslational modification and intracellular transport of a trypanosome variant surface glycoprotein. *The Journal of Cell Biology* **103**:255-263 <https://doi.org/10.1083/jcb.103.1.255>
5. Barcons-Simon A., Carrington M., Siegel T. N (2023) Decoding the impact of nuclear organization on antigenic variation in parasites. *Nature Microbiology* **8**:1408-1418 <https://doi.org/10.1038/s41564-023-01424-9>

6. Beneke T., Madden R., Makin L., Valli J., Sunter J., Gluenz E (2017) A CRISPR Cas9 high-throughput genome editing toolkit for kinetoplastids. *Royal Society Open Science* **4**:170095 <https://doi.org/10.1098/rsos.170095>
7. Biebinger S., Helfert S., Steverding D., Ansorge I., Clayton C (2003) Impaired dimerization and trafficking of ESAG6 lacking a glycosyl-phosphatidylinositol anchor. *Molecular and Biochemical Parasitology* **132**:93-96 <https://doi.org/10.1016/j.molbiopara.2003.07.005>
8. Billington K., Halliday C., Madden R., Dyer P., Barker A. R., Moreira-Leite F. F., Carrington M., Vaughan S., Hertz-Fowler C., Dean S., et al. (2023) Genome-wide subcellular protein map for the flagellate parasite *Trypanosoma brucei*. *Nature Microbiology* **8**:533-547 <https://doi.org/10.1038/s41564-022-01295-6>
9. Bolger A. M., Lohse M., Usadel B (2014) Trimmomatic: a flexible trimmer for Illumina sequence data. *Bioinformatics* **30**:2114-2120 <https://doi.org/10.1093/bioinformatics/btu170>
10. Bonhivers M., Nowacki S., Landrein N., Robinson D. R (2008) Biogenesis of the trypanosome endo-exocytotic organelle is cytoskeleton mediated. *PLoS Biology* **6**:e105 <https://doi.org/10.1371/journal.pbio.0060105>
11. Chaudhri M., Steverding D., Kittelberger D., Tjia S., Overath P (1994) Expression of a glycosylphosphatidylinositol-anchored *Trypanosoma brucei* transferrin-binding protein complex in insect cells. *Proceedings of the National Academy of Sciences of the United States of America* **91**:6443-6447 <https://doi.org/10.1073/pnas.91.14.6443>
12. Cook A. D., Carrington M., Higgins M. K (2024) Molecular mechanism of complement inhibition by the trypanosome receptor ISG65. *eLife* **12** <https://doi.org/10.7554/eLife.88960>
13. Duncan S. M., Carbajo C. G., Nagar R., Zhong Q., Breen C., Ferguson M. A. J., Tiengwe C (2024) Generation of a bloodstream form *Trypanosoma brucei* double glycosyltransferase null mutant competent in receptor-mediated endocytosis of transferrin. *PLoS Pathogens* **20**:e1012333 <https://doi.org/10.1371/journal.ppat.1012333>
14. Engstler M., Pfohl T., Herminghaus S., Boshart M., Wiegertjes G., Heddergott N., Overath P (2007) Hydrodynamic flow-mediated protein sorting on the cell surface of trypanosomes. *Cell* **131**:505-515 <https://doi.org/10.1016/j.cell.2007.08.046>
15. Engstler M., Thilo L., Weise F., Grünfelder C. G., Schwarz H., Boshart M., Overath P (2004) Kinetics of endocytosis and recycling of the GPI-anchored variant surface glycoprotein in *Trypanosoma brucei*. *Journal of Cell Science* **117**:1105-1115 <https://doi.org/10.1242/jcs.00938>
16. Gadelha C., Rothery S., Morphew M., McIntosh J. R., Severs N. J., Gull K (2009) Membrane domains and flagellar pocket boundaries are influenced by the cytoskeleton in African trypanosomes. *Proceedings of the National Academy of Sciences of the United States of America* **106**:17425-17430 <https://doi.org/10.1073/pnas.0909289106>
17. Gadelha C., Zhang W., Chamberlain J. W., Chait B. T., Wickstead B., Field M. C (2015) Architecture of a Host-Parasite Interface: Complex Targeting Mechanisms Revealed Through Proteomics. *Molecular & Cellular Proteomics : MCP* **14**:1911-1926 <https://doi.org/10.1074/mcp.M114.047647>
18. Gerrits H., Musmann R., Bitter W., Kieft R., Borst P (2002) The physiological significance of transferrin receptor variations in *Trypanosoma brucei*. *Molecular and Biochemical Parasitology* **119**:237-247 [https://doi.org/10.1016/S0166-6851\(01\)00417-0](https://doi.org/10.1016/S0166-6851(01)00417-0)
19. Gilabert Carbajo C., Cornell L. J., Madbouly Y., Lai Z., Yates P. A., Tinti M., Tiengwe C (2021) Novel aspects of iron homeostasis in pathogenic bloodstream form *Trypanosoma brucei*. *PLoS Pathogens* **17**:e1009696 <https://doi.org/10.1371/journal.ppat.1009696>
20. Gíslason M. H., Nielsen H., Almagro Armenteros J. J., Johansen A. R (2021) Prediction of GPI-anchored proteins with pointer neural networks. *Current Research in Biotechnology* **3**:6-13 <https://doi.org/10.1016/j.crbiot.2021.01.001>
21. Hertz-Fowler C., Figueiredo L. M., Quail M. A., Becker M., Jackson A., Bason N., Brooks K., Churcher C., Fahkro S., Goodhead I., et al. (2008) Telomeric expression sites are highly conserved in *Trypanosoma brucei*. *PloS One* **3**:e3527 <https://doi.org/10.1371/journal.pone.0003527>

22. Higgins M. K., Tkachenko O., Brown A., Reed J., Raper J., Carrington M (2013) Structure of the trypanosome haptoglobin-hemoglobin receptor and implications for nutrient uptake and innate immunity. *Proceedings of the National Academy of Sciences of the United States of America* **110**:1905-1910 <https://doi.org/10.1073/pnas.1214943110>
23. Hirumi H., Hirumi K (1989) Continuous cultivation of *Trypanosoma brucei* blood stream forms in a medium containing a low concentration of serum protein without feeder cell layers. *The Journal of Parasitology* **75**:985-989
24. Hsia R., Beals T., Boothroyd J. C (1996) Use of chimeric recombinant polypeptides to analyse conformational, surface epitopes on trypanosome variant surface glycoproteins. *Molecular Microbiology* **19**:53-63 <https://doi.org/10.1046/j.1365-2958.1996.351878.x>
25. Kabiri M., Steverding D (2000) Studies on the recycling of the transferrin receptor in *Trypanosoma brucei* using an inducible gene expression system. *European Journal of Biochemistry* **267**:3309-3314 <https://doi.org/10.1046/j.1432-1327.2000.01361.x>
26. Kakuta K., Orino K., Yamamoto S., Watanabe K (1997) High levels of ferritin and its iron in fetal bovine serum. *Comparative Biochemistry and Physiology Part A: Physiology* **118**:165-169 [https://doi.org/10.1016/s0300-9629\(96\)00403-3](https://doi.org/10.1016/s0300-9629(96)00403-3)
27. Lane-Serff H., MacGregor P., Lowe E. D., Carrington M., Higgins M. K (2014) Structural basis for ligand and innate immunity factor uptake by the trypanosome haptoglobin-haemoglobin receptor. *eLife* **3**:e05553 <https://doi.org/10.7554/eLife.05553>
28. Lane-Serff H., MacGregor P., Peacock L., Macleod O. J., Kay C., Gibson W., Higgins M. K., Carrington M (2016) Evolutionary diversification of the trypanosome haptoglobin-haemoglobin receptor from an ancestral haemoglobin receptor. *eLife* **5** <https://doi.org/10.7554/eLife.13044>
29. Leung K. F., Riley F. S., Carrington M., Field M. C (2011) Ubiquitylation and developmental regulation of invariant surface protein expression in trypanosomes. *Eukaryotic Cell* **10**:916-931 <https://doi.org/10.1128/EC.05012-11>
30. Ligtenberg M. J., Bitter W., Kieft R., Steverding D., Janssen H., Calafat J., Borst P (1994) Reconstitution of a surface transferrin binding complex in insect form *Trypanosoma brucei*. *The EMBO Journal* **13**:2565-2573 <https://doi.org/10.1002/j.1460-2075.1994.tb06546.x>
31. MacGregor P., Gonzalez-Munoz AL, Jobe F, Taylor MC, Rust S, Sandercock AM, Macleod OJS, Van Bocxlaer K, Francisco A F, D'Hooge F, *et al.* (2019) A single dose of antibody-drug conjugate cures a stage 1 model of African trypanosomiasis. *PLoS Neglected Tropical Diseases* **13**:e0007373 <https://doi.org/10.1371/journal.pntd.0007373>
32. Macleod O. J. S., Bart J.-M., MacGregor P., Peacock L., Savill N. J., Hester S., Ravel S., Sunter J. D., Trevor C., Rust S., *et al.* (2020) A receptor for the complement regulator factor H increases transmission of trypanosomes to tsetse flies. *Nature Communications* **11**:1326 <https://doi.org/10.1038/s41467-020-15125-y>
33. Macleod O. J. S., Cook A. D., Webb H., Crow M., Burns R., Redpath M., Seisenberger S., Trevor C. E., Peacock L., Schwede A., *et al.* (2022) Invariant surface glycoprotein 65 of *Trypanosoma brucei* is a complement C3 receptor. *Nature Communications* **13**:5085 <https://doi.org/10.1038/s41467-022-32728-9>
34. Maier A., Steverding D (2008) Expression and purification of non-glycosylated *Trypanosoma brucei* transferrin receptor in insect cells. *Experimental Parasitology* **120**:205-207 <https://doi.org/10.1016/j.exppara.2008.07.004>
35. Mikkelsen J. H., Stødkilde K., Jensen M. P., Hansen A. G., Wu Q., Lorentzen J., Graversen J. H., Andersen G. R., Fenton R. A., Etzerodt A., *et al.* (2024) *Trypanosoma brucei* Invariant Surface Glycoprotein 75 Is an Immunoglobulin Fc Receptor Inhibiting Complement Activation and Antibody-Mediated Cellular Phagocytosis. *Journal of Immunology* **212**:1334 <https://doi.org/10.4049/jimmunol.2300862>
36. Mugnier M. R., Cross G. A. M., Papavasiliou F. N (2015) The in vivo dynamics of antigenic variation in *Trypanosoma brucei*. *Science* **347**:1470-1473 <https://doi.org/10.1126/science.aaa4502>

37. **Mussmann R.**, Engstler M., Gerrits H., Kieft R., Toaldo C. B., Onderwater J., Koerten H., van Luenen H. G. A. M., Borst P. (2004) Factors affecting the level and localization of the transferrin receptor in *Trypanosoma brucei*. *The Journal of Biological Chemistry* **279**:40690-40698 <https://doi.org/10.1074/jbc.M404697200>
38. **Mussmann R.**, Janssen H., Calafat J., Engstler M., Ansorge I., Clayton C., Borst P (2003) The expression level determines the surface distribution of the transferrin receptor in *Trypanosoma brucei*. *Molecular Microbiology* **47**:23-35 <https://doi.org/10.1046/j.1365-2958.2003.03245.x>
39. **Pays E.**, Lips S., Nolan D., Vanhamme L., Pérez-Morga D (2001) The VSG expression sites of *Trypanosoma brucei*: multipurpose tools for the adaptation of the parasite to mammalian hosts. *Molecular and Biochemical Parasitology* **114**:1-16 [https://doi.org/10.1016/s0166-6851\(01\)00242-0](https://doi.org/10.1016/s0166-6851(01)00242-0)
40. **Pinger J.**, Nešić D., Ali L., Aresta-Branco F., Lilic M., Chowdhury S., Kim H.-S., Verdi J., Raper J., Ferguson M. A. J., *et al.* (2018) African trypanosomes evade immune clearance by O-glycosylation of the VSG surface coat. *Nature Microbiology* **3**:932-938 <https://doi.org/10.1038/s41564-018-0187-6>
41. **Redpath M. B.**, Windle H., Nolan D., Pays E., Voorheis H. P., Carrington M (2000) ESAG11, a new VSG expression site-associated gene from *Trypanosoma brucei*. *Molecular and Biochemical Parasitology* **111**:223-228 [https://doi.org/10.1016/s0166-6851\(00\)00305-4](https://doi.org/10.1016/s0166-6851(00)00305-4)
42. **Roggy J. L.**, Bangs J. D (1999) Molecular cloning and biochemical characterization of a VCP homolog in African trypanosomes. *Molecular and Biochemical Parasitology* **98**:1-15 [https://doi.org/10.1016/s0166-6851\(98\)00114-5](https://doi.org/10.1016/s0166-6851(98)00114-5)
43. **Salmon D.**, Geuskens M., Hanocq F., Hanocq-Quertier J., Nolan D., Ruben L., Pays E (1994a) A novel heterodimeric transferrin receptor encoded by a pair of VSG expression site-associated genes in *T. brucei*. *Cell* **78**:75-86 [https://doi.org/10.1016/0092-8674\(94\)90574-6](https://doi.org/10.1016/0092-8674(94)90574-6)
44. **Salmon D.**, Geuskens M., Hanocq F., Hanocq-Quertier J., Nolan D., Ruben L., Pays E (1994b) A novel heterodimeric transferrin receptor encoded by a pair of VSG expression site-associated genes in *T. brucei*. *Cell* **78**:75-86 [https://doi.org/10.1016/0092-8674\(94\)90574-6](https://doi.org/10.1016/0092-8674(94)90574-6)
45. **Schwartz K. J.**, Peck R. F., Tazeh N. N., Bangs J. D (2005) GPI valence and the fate of secretory membrane proteins in African trypanosomes. *Journal of Cell Science* **118**:5499-5511 <https://doi.org/10.1242/jcs.02667>
46. **Schwede A.**, Jones N., Engstler M., Carrington M (2011) The VSG C-terminal domain is inaccessible to antibodies on live trypanosomes. *Molecular and Biochemical Parasitology* **175**:201-204 <https://doi.org/10.1016/j.molbiopara.2010.11.004>
47. **Steverding D.**, Stierhof Y. D., Chaudhri M., Ligtenberg M., Schell D., Beck-Sickinger A. G., Overath P (1994a) ESAG 6 and 7 products of *Trypanosoma brucei* form a transferrin binding protein complex. *European Journal of Cell Biology* **64**:78-87
48. **Steverding D.**, Stierhof Y. D., Chaudhri M., Ligtenberg M., Schell D., Beck-Sickinger A. G., Overath P (1994b) ESAG 6 and 7 products of *Trypanosoma brucei* form a transferrin binding protein complex. *European Journal of Cell Biology* **64**:78-87
49. **Steverding D.**, Stierhof Y. D., Fuchs H., Tauber R., Overath P (1995) Transferrin-binding protein complex is the receptor for transferrin uptake in *Trypanosoma brucei*. *The Journal of Cell Biology* **131**:1173-1182 <https://doi.org/10.1083/jcb.131.5.1173>
50. **Tiengwe C.**, Bush P. J., Bangs J. D (2017) Controlling transferrin receptor trafficking with GPI-valence in bloodstream stage African trypanosomes. *PLoS Pathogens* **13**:e1006366 <https://doi.org/10.1371/journal.ppat.1006366>
51. **Trevor C. E.**, Gonzalez-Munoz A. L., Macleod O. J. S., Woodcock P. G., Rust S., Vaughan T. J., Garman E. F., Minter R., Carrington M., Higgins M. K (2019) Structure of the trypanosome transferrin receptor reveals mechanisms of ligand recognition and immune evasion. *Nature Microbiology* **4**:2074-2081 <https://doi.org/10.1038/s41564-019-0589-0>
52. **van Luenen H. G. A. M.**, Kieft R., Mussmann R., Engstler M., ter Riet B., Borst P (2005) Trypanosomes change their transferrin receptor expression to allow effective uptake of host transferrin. *Molecular Microbiology* **58**:151-165 <https://doi.org/10.1111/j.1365-2958.2005.04831.x>

53. Vanhollebeke B., De Muylder G., Nielsen M. J., Pays A., Tebabi P., Dieu M., Raes M., Moestrup S. K., Pays E. (2008) A haptoglobin-hemoglobin receptor conveys innate immunity to *Trypanosoma brucei* in humans. *Science* **320**:677-681 <https://doi.org/10.1126/science.1156296>
54. Ziegelbauer K., Multhaup G., Overath P (1992) Molecular characterization of two invariant surface glycoproteins specific for the bloodstream stage of *Trypanosoma brucei*. *The Journal of Biological Chemistry* **267**:10797-10803
55. Ziegelbauer K., Overath P (1992) Identification of invariant surface glycoproteins in the bloodstream stage of *Trypanosoma brucei*. *The Journal of Biological Chemistry* **267**:10791-10796
56. Ziegelbauer K., Overath P (1993) Organization of two invariant surface glycoproteins in the surface coat of *Trypanosoma brucei*. *Infection and Immunity* **61**:4540-4545 <https://doi.org/10.1128/iai.61.11.4540-4545.1993>

Peer reviews

Reviewer #1 (Public review):

Summary:

An interesting manuscript from the Carrington lab is presented investigating the behavior of single vs double GPI-anchored nutrient receptors in bloodstream form (BSF) *T. brucei*. These include the transferrin receptor (TfR), the HpHb receptor (HpHbR), and the factor H receptor (FHR). The central question is why these critical proteins are not targeted by host acquired immunity. It has generally been thought that they are sequestered in the flagellar pocket (FP), where they are subject to rapid endocytosis - any Ab:receptor complexes would be rapidly removed from the cell surface. This manuscript challenges that assumption by showing that these receptors can be found all over the outer cell body and flagella surfaces - if one looks in an appropriate manner (rapid direct fixation in culture media).

Strengths and weaknesses:

(1) The presence of a second ESAG6 gene in the BES7 expression site was noted in the previous review. This is now noted and discussed appropriately in the current version.

(2) Surface binding studies: The ability of cells to bind tagged-Tf while in complete media was challenged and it was suggested that classic competition studies be performed to validate saturable ligand binding. This has been done now and the results confirm that this is so. A reasonable discussion of the results is presented.

(3) Variable TfR expression in different BESs: The claim that specific ES environment is the dominant factor controlling TfR expression levels was challenged in that the presented results could be due to technical issues. RNA seq has now been performed confirming that the differences in TfR abundance is indeed directly related to mRNA levels

(4) Surface immuno-localization of receptors: In regard to the novel immunofluorescence (direct fixation) methodology used to demonstrate TfR on the cell surface the authors were asked if they had attempted more traditional methods that involve centrifugation/washing. These data are now provided (Fig S5) and do indicate that centrifugation does reduce signal, likely due to shedding and/or internalization during the procedure. Nevertheless, significant signal is present after centrifugation leaving the issue of why others have never detected significant surface TfR.

These responses address all the major concerns with the original submission and a greatly improved manuscript is now submitted.

<https://doi.org/10.7554/eLife.107191.2.sa2>

Reviewer #2 (Public review):

The revised data support the conclusion that methodological differences can influence apparent receptor localization. However, key claims regarding functional surface engagement of TfR and hydrodynamic clearance remain based largely on indirect evidence and model-based interpretation. These conclusions should therefore be phrased more cautiously.

I thank the authors for their careful rebuttal and the additional experiments included in the revised manuscript. The new fixation comparisons and transferrin competition assays substantially strengthen the technical basis of the study and address several of the original concerns.

However, some conclusions remain more inferential than directly supported by the data. While the fixation and washing controls demonstrate that methodology influences apparent TfR localisation, they do not directly establish that previous protocols quantitatively redistribute surface TfR into the flagellar pocket. Statements implying such redistribution should therefore be phrased more cautiously.

Similarly, the added transferrin binding controls argue against non-specific interactions, but functional engagement of surface-exposed TfR in intact bloodstream-form parasites remains supported mainly by indirect evidence. The proposed explanation involving rapid on/off rates and newly arriving receptors is plausible but should be more clearly identified as an inference.

<https://doi.org/10.7554/eLife.107191.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

*An interesting manuscript from the Carrington lab is presented investigating the behavior of single vs double GPI-anchored nutrient receptors in bloodstream form (BSF) *T. brucei*. These include the transferrin receptor (TfR), the HpHb receptor (HpHbR), and the factor H receptor (FHR). The central question is why these critical proteins are not targeted by host-acquired immunity. It has generally been thought that they are sequestered in the flagellar pocket (FP), where they are subject to rapid endocytosis - any Ab:receptor complexes would be rapidly removed from the cell surface. This manuscript challenges that assumption by showing that these receptors can be found all over the outer cell body and flagella surfaces, if one looks in an appropriate manner (rapid direct fixation in culture media).*

The main part of the manuscript focuses on TfR, typically a GPI1 heterodimer of very similar E6 (GPI anchored) and E7 (truncated, no GPI) subunits. These are expressed coordinately from 15 telomeric expression sites (BES), of which only one can be transcribed at a time. The authors identify a native E6:E7 pair in BES7 in which E7 is not truncated and therefore forms a GPI2 heterodimer. By in situ genetic manipulation, they generate two different sets of GPI1:GPI2 TfR combinations expressed from two different BESs (BES1 and BES7). Comparative analyses of these receptors form the bulk of the data.

The main findings are:

- (1) Both GPI1 and GPI2 TfR can be found on the cell body/flagellar surface.
- (2) Both are functional for Tf binding and uptake.
- (3) GPI2 TfR is expressed at ~1.5x relative to GPI1 TfR
- (4) Ultimate TfR expression level (protein) is dependent on the BES from which it is expressed.

Most of these results are quite reasonably explained in light of the hydrodynamic flow model of the Engstler lab and the GPI valence model of the Bangs lab. Additional experiments, again by rapid fixation, with HpHbR and FHR, show that these GPI1 receptors can also be seen on the cell surface, in contrast to published localizations.

It is quite interesting that the authors have identified a native GPI2 TfR. However, essentially all of the data with GPI2 TfR are confirmatory for the prior, more detailed studies of Tiengwe et al. (2017). That said, the suggestion that GPI2 was the ancestral state makes good evolutionary sense, and begs the question of why trypanosomes prefer GPI1 TfR in 14 of 15 ESs (i.e., what is the selection pressure?)

Strengths and weaknesses:

- (1) BES7 TfR subunit genes (BES7_Tb427v10): There are actually three (in order 5'3'): E7gpi, E6.1 and E6.2. E6.1 and E6.2 have a single nucleotide difference. This raises the issue of coordinate expression. If overall levels of E6 (2 genes) are not down-regulated to match E7 (1 gene), this will result in a 2x excess of E6 subunits. The most likely fate of these is the formation of non-functional GPI2 homodimers on the cell surface, as shown in Tiengwe et al. (2017), which will contribute to the elevated TfR expression seen in BES7.

We would like to thank the reviewer for pointing out that there are two ESAG6 genes in BES7, we had relied on the publicly available annotation and should have known better.

For transferrin expression levels, see the discussion in response to reviewer 1 point 3 below

- (2) Surface binding studies: This is the most puzzling aspect of the entire manuscript. That surface GPI2 TfR should be functional for Tf binding and uptake is not surprising, as this has already been shown by Tiengwe et al. (2017), but the methodology for this assay raises important questions. First, labeled Tf is added at 500 nM to live cells in complete media containing 2.5 μ M unlabeled Tf - a 5x excess. It is difficult to see how significant binding of labeled TfR could occur in as little as 15 seconds under these conditions.

The k_{on} for transferrin is very rapid (BES1 TfR / bovine transferrin at pH7.4 = $4.5 \times 10^5 \text{ M}^{-1}\text{s}^{-1}$ (Trevor et al., 2019) and binding would occur to unoccupied receptors within 15 sec. The k_{off} is also fast (BES1 TfR / bovine transferrin at pH7.4 = $3.6 \times 10^{-2} \text{ s}^{-1}$ (Trevor et al., 2019) and there would be exchange of transferrin within the time taken for endocytosis. These values are *in vitro* with purified proteins, the *in vivo* values may be affected by the VSG coat.

The failure to bind canine transferrin (Supp. Figure 4B) acts as a control for specificity of the interaction.

We have now performed a competition experiment as an additional control; cells in culture were supplemented with: A, 0.5 μ M labelled transferrin; B, 0.5 μ M labelled and 2.5 μ M unlabelled transferrin; C, 0.5 μ M labelled and 5 μ M unlabelled transferrin, fixed after 60 s and visualised by fluorescence microscopy (Figure S4C). There was effective competition and greatly reduced binding of transferrin was seen in the presence of a 10-fold excess of unlabelled. We would like to thank the reviewer for suggesting this experiment.

Second, Tiengwe et al. (2017) found that trypanosomes taken directly from culture could not bind labeled Tf in direct surface labelling experiments. To achieve binding, it was necessary to first culture cells in serum-free media for a sufficient time to allow new unligated TfR to be synthesized and transported to the surface. This result suggests that essentially all surface TfR is normally ligated and unavailable to the added probe.

As part of the preliminary experiments for this paper we found that centrifugation followed by resuspension in either complete or serum free (but 1% BSA) medium resulted in a reduction in total cellular TfR and determined by western blotting. We have now included this experiment (Figure S4D). The inference from this experiment is that centrifugation and subsequently incubation will have an effect on receptor detection and endocytosis rates for a discrete time period.

The amount of binding of labelled transferrin to cells in culture will depend on the specific activity of the labelled transferrin. This reasoning was behind the use of 0.5 μ M labelled transferrin when roughly 1 in 6 molecules in the culture medium are labelled and there was only a small effect on the overall concentration of transferrin.

Third, the authors have themselves argued previously, based on binding affinities, that all surface-exposed TfR is likely ligated in a natural setting (DOI:10.1002/bies.202400053). Could the observed binding actually be non-specific due to the high levels of fixative used?

The absence of binding/uptake of canine transferrin argues against a non-specific interaction. In our previous publication, we did not pay enough attention to the on and off rates which allow for a degree of exchange and, here, TfR newly appearing on the cell surface has a 1 in 6 chance of binding a labelled transferrin.

(3) Variable TfR expression in different BESs: It appears that native TfR is expressed at higher levels from BES7 compared to BES1, and even more so when compared to BES3. This raises the possibility that the anti-TfR used in these experiments has differential reactivity with the three sets of TfRs. The authors discount this possibility due to the overall high sequence similarities of E6s and E7s from the various ESs. However, their own analyses show that the BES1, BES3, and BES7 TfRs are relatively distal to each other in the phylogenetic trees, and this Reviewer strongly suspects that the apparent difference in expression is due to differential reactivity with the anti-TfR used in this work. In the grand scheme, this is a minor issue that does not impact the other major conclusions concerning TfR localization and function, nor the behavior of HpHbR and FHR. However, the authors make very strong conclusions about the role of BESs in TfR expression levels, even claiming that it is the 'dominant determinant' (line 189).

This point is valid but exceptionally difficult to address at the protein level. As an orthogonal approach, we performed RNAseq analysis of the 'wild type' BES1, BES3, and BES7 cell lines to determine whether differences in receptor mRNA levels were consistent with the proposed difference in protein levels (Table S1). The analysis showed total ESAG6/7 mRNA levels to vary in a similar manner to the protein estimates with BES3 < BES1 < BES7 providing support for the differences in protein levels.

The strongest evidence for the expression site determining the TfR level is the comparison of the cell lines in which the VSG were exchanged. This had no effect on TfR levels and so there is no evidence that the identity of the VSG alters TfR expression.

(4) Surface immuno-localization of receptors: These experiments are compelling and useful to the field. To explain the difference with essentially all prior studies, the authors suggest that typical fixation procedures allow for clearance of receptor:ligand complexes by hydrodynamic flow due to extended manipulation prior to fixation (washing steps).

Despite the fact that these protocols typically involve ice-cold physiological buffers that minimize membrane mobility, this is a reasonable possibility. Have the authors challenged their hypothesis by testing more typical protocols themselves? Other contributing factors that could play a role are the use of deconvolution, which tends to minimize weak signals, and also the fact that investigators tend to discount weak surface signals as background relative to stronger internal signals.

We have added preliminary experiments that compared fixation protocols in two parts. First the effect on TfR levels of washing and resuspending cells discussed above (Figure S4D), and second how different fixation protocols alter apparent TfR immunolocalisation (Supp Figure S5A-B). The comparison shows that both the absence of glutaraldehyde and the use of washing alters the outcome.

(5) Shedding: A central aspect of the GPI valence model (Schwartz et al., 2005, Tiengwe et al., 2017) is that GPI1 reporters that reach the cell body surface are shed into the media because a single dimyristoylglycerol-containing GPI anchor does not stably associate with biological membranes. As the authors point out, this is a major factor contributing to higher steady-state levels of cell-associated GPI2 TfR relative to GPI1 TfR. Those studies also found that the size/complexity of the attached protein correlated inversely with shedding, suggesting exit from the flagellar pocket as a restricting factor in cell body surface localization. The amount of newly synthesized TfR shed into the media was ~5%, indicating that very little actually exits the FP to the outer surface. In this regard, is it possible to know the overall ratio of cell surface:FP:endosomal localized receptors? Could these data not be 'harvested' from the 3D structural illumination imaging?

A ratio could be determined but we did not do this as it would only be valid if the antibody has equal access to the internal TfR in a diluted VSG environment and the external VSG embedded in a densely packed and cross-linked VSG layer. As such, we would have no confidence in the accuracy of any estimate.

Reviewer #2 (Public review):

The work has significant implications for understanding immune evasion and nutrient uptake mechanisms in trypanosomes.

While the experimental rigor is commendable, revisions are needed to clarify methodological limitations and to broaden the discussion of functional consequences.

The authors argue that prior studies missed surface-localized TfR due to harsh washing/fixation (e.g., methanol). While this is plausible, additional evidence would strengthen the claim.

Preliminary experiments that compared fixation protocols are now included to show that method affects outcome.

It remains unclear how centrifugation steps of various lengths (as in previous publications) can equally and quantitatively redistribute TfR into the flagellar pocket. If this were the case, it should be straightforward for the authors to test this experimentally.

Not aware of previous studies that demonstrate equal and quantitative redistribution to the flagellar pocket. In previous reports, there is variation in cell surface/flagellar pocket localisation depending on expression levels, for example (Mussmann et al., 2003) (Mussmann et al., 2004), it's worth noting that the increase in TfR expression in these papers is similar to the difference in the cell lines used here. In addition, most report the presence of TfR in endosomal compartments. In the experiments here, there are cells where the majority of signal from labelled transferrin is present in the flagellar pocket and the argument is that this

is a stage of a continuous process in which the receptor picks up a transferrin on the cell surface and is swept towards the pocket.

If TfR is distributed over the cell surface, live-cell imaging with fluorescent transferrin should be performed as a control. Modern detection limits now reach the single-molecule level, and transient immobilization of live trypanosomes has been established, which would exclude hydrodynamic surface clearance as a confounding factor.

This is non-trivial and is a longer-term aim. The immobilisation involves significant manipulation of the cells prior to restraining.

In most images, TfR is not evenly distributed on the surface but rather appears punctate. Could this reflect localization to membrane domains? Immuno-EM with high-pressure frozen parasites could resolve this question and is relatively straightforward.

There is a non-uniform appearance in the super-resolution images for both TfR and FHR. We cannot distinguish whether this represents random variation in receptor density over the cell surface or results from a biological phenomenon. Whatever the cause, the experiments showed unambiguous cell surface localisation.

The authors might consider discussing whether differences in parasite life cycle stages (procyclic versus bloodstream forms) or culture conditions (e.g., cell density) affect localization. The developmentally regulated retention of GPI-anchored procyclin in the flagellar pocket might be worth mentioning.

The aim of this paper was to determine the localisation of receptors in proliferating bloodstream form trypanosomes in culture. TfR and HpHbR are not expressed in insect stages in culture. FHR is expressed in insect stages and is present all over the cell surface (Macleod et al., 2020). A procyclin-based reporter was distributed over the whole cell surface in one report (Schwartz et al. 2005). In other reports, the retention of procyclin in the flagellar pocket of proliferating bloodstream forms is probably dependent on structure/sequence as other single GPI-anchored proteins, such as FHR (Macleod et al., 2020) and GPI-anchored sfGFP (Martos-Esteban et al., 2022) can access the surface.

References:

- MacGregor, P., Gonzalez-Munoz, A. L., Jobe, F., Taylor, M. C., Rust, S., Sandercock, A. M., Macleod, O. J. S., Van Bocxlaer, K., Francisco, A. F., D'Hooge, F., Tiberghien, A., Barry, C. S., Howard, P., Higgins, M. K., Vaughan, T. J., Minter, R., & Carrington, M. (2019). A single dose of antibody-drug conjugate cures a stage 1 model of African trypanosomiasis. *PLoS Neglected Tropical Diseases*, 13(5), e0007373. <https://doi.org/10.1371/journal.pntd.0007373>
- Macleod, O. J. S., Bart, J.-M., MacGregor, P., Peacock, L., Savill, N. J., Hester, S., Ravel, S., Sunter, J. D., Trevor, C., Rust, S., Vaughan, T. J., Minter, R., Mohammed, S., Gibson, W., Taylor, M. C., Higgins, M. K., & Carrington, M. (2020). A receptor for the complement regulator factor H increases transmission of trypanosomes to tsetse flies. *Nature Communications*, 11(1), 1326. <https://doi.org/10.1038/s41467-020-15125-y>
- Martos-Esteban, A., Macleod, O. J. S., Maudlin, I., Kalogeropoulos, K., Jürgensen, J. A., Carrington, M., & Laustsen, A. H. (2022). Black-necked spitting cobra (*Naja nigricollis*) phospholipases A2 may cause *Trypanosoma brucei* death by blocking endocytosis through the flagellar pocket. *Scientific Reports*, 12(1), 6394. <https://doi.org/10.1038/s41598-02210091-5>
- Mussmann, R., Engstler, M., Gerrits, H., Kieft, R., Toaldo, C. B., Onderwater, J., Koerten, H., van Luenen, H. G. A. M., & Borst, P. (2004). Factors affecting the level and localization of the

transferrin receptor in *Trypanosoma brucei*. The Journal of Biological Chemistry, 279(39), 40690–40698. <https://doi.org/10.1074/jbc.M404697200>

Mussmann, R., Janssen, H., Calafat, J., Engstler, M., Ansorge, I., Clayton, C., & Borst, P. (2003). The expression level determines the surface distribution of the transferrin receptor in *Trypanosoma brucei*. Molecular Microbiology, 47(1), 23–35. <https://doi.org/10.1046/j.13652958.2003.03245.x>

Schwartz, K. J., Peck, R. F., Tazeh, N. N., & Bangs, J. D. (2005). GPI valence and the fate of secretory membrane proteins in African trypanosomes. Journal of Cell Science, 118(Pt 23), 5499–5511. <https://doi.org/10.1242/jcs.02667>

Trevor, C. E., Gonzalez-Munoz, A. L., Macleod, O. J. S., Woodcock, P. G., Rust, S., Vaughan, T. J., Garman, E. F., Minter, R., Carrington, M., & Higgins, M. K. (2019). Structure of the trypanosome transferrin receptor reveals mechanisms of ligand recognition and immune evasion. Nature Microbiology, 4(12), 2074–2081. <https://doi.org/10.1038/s41564-019-0589-0>

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Major Recommendations:

(1) 2 E6 gene in BES7s: This does not affect the overall conclusions, but the text should be modified to reflect the existence of the second gene, and to discuss the ramifications.

This has been corrected

(2) Surface binding studies: To clarify this issue, two experimental approaches are strongly recommended. First: additional excess unlabelled Tf should be added. If binding is truly receptor-mediated, it must by definition be saturable at some experimentally achievable level. Second: TfR expression should be abrogated by RNAi silencing to show that binding is TfR-dependent. Without some validation of specific binding by one or both of these approaches, these counter-intuitive results must be questioned.

The excess unlabelled transferrin experiment is now included (we would like to thank the reviewer for this suggestion). The absence of binding of canine transferrin provides strong evidence for the specificity.

(3) Variable TfR expression in different BESs: To make such claims, quantitative RTPCR should be performed with conserved primers to assess the actual relative expression at the transcriptional level. Absent this, the claims should be eliminated, or at the very least greatly tempered.

This has been done using an RNAseq analysis.

(4) Surface immuno-localization of receptors: An example of discounting weak signals as background can be seen in Figure 8 of Duncan et al. (2024). It has also been shown that at least one other GPI1 reporter (procyclin) is readily detected on the outer cell surface under ectopic expression in BSF trypanosomes (Schwartz et al., 2005) using typical fixation procedures. This could be cited, and the authors could discuss the fact that procyclin is not a receptor and may not be susceptible to hydrodynamic drag.

Yes

Minor issues:

(1) Fully appreciating the data presented requires an understanding of the hydrodynamic flow and GPI valence models of the Engstler and Bangs labs, respectively.

For the uninitiated, it might perhaps be useful to include brief summaries of each in the Introduction.

Added to the introduction

(2) Lines 110-112: ISG65 and ISG75 both have strong localizations in endosomal compartments. This should be noted with citation of any of the work from the Field lab.

Added

(3) Lines 121-132: This passage presents the role of GPI anchors (1 vs 2) in a rather digital manner (in or out). Schwartz et al (2005) present a much more nuanced view of what is likely taking place. This is one reason summaries of hydrodynamic flow and GPI valence would be helpful.

Modified

(4) Lines 182-184: The increased size of GPI-anchored E7 is in part due to the presence of the GPI itself, as the authors state, but there are also 24 additional amino acid residues in this protein that contribute.

Modified

(5) Lines 212-214: Do $p > 0.95$ and $p > 0.99$ indicate statistical significance? This must be a typo.

Thank you, corrected

(6) Lines 218-219: The better references documenting GPI number in regard to turnover/shedding are Schwartz et al. 2005 and Tiengwe et al. 2017.

Changed

(7) Line 241 and Figures 3, 4, and 6: The transverse sections add little to the presentation. That there is signal variation in all dimensions is readily apparent from the images themselves, and similar profiles would be obtained regardless of the transect. Was there some process/rationale in the selection of the individual transects intended to make a broader point? If so, a description of the process should be provided.

The point was to show that the signal had a pattern consistent with plasma membrane (two distal peaks) as opposed to cytoplasm (single central peak). As such, we think it is important.

(8) Lines 582-596: Methodology for quantitation of cellular fluorescent signals should be provided.

Has been expanded

Reviewer #2 (Recommendations for the authors):

(1) As a less critical but still useful control, antibody accessibility assays on live versus fixed parasites could test whether VSG coats limit detection.

This could only be quantified by using a range of monoclonal antibodies which are not available.

(2) The rapid transferrin uptake (15-60 seconds) could reflect fast endocytic recycling rather than stable surface residency. A pulse-chase experiment tracking receptor movement would clarify this (though I acknowledge that this is technically challenging).

We agree that endocytic recycling is probably the main source of unoccupied TfR on the cell surface. It is hard to see how the pulse chase experiment could be performed without centrifugation which will affect the outcome – see above.

| (3) *Statistical and quantitative reporting*

Added as Table S2- S4

| (4) *Report confidence intervals (e.g., for fluorescence intensity comparisons in Figure 3B) to contextualize claims of "no significant difference."*

We do not claim ‘no significant difference’ and the SD overlap due to a high level of variation in the population

| (5) *Specify the number of biological replicates and cells analyzed per condition in the figure legends.*

Added

| (6) *The study notes that surface-exposed receptors avoid antibody detection, but does not explore how.*

We don’t claim that receptors avoid detection and have published evidence to the contrary. The cell has evolved mechanisms to reduce/minimise the effect of antibody binding.

| (7) *Comparing antibody binding to TfR in VSG221 versus VSG224 coats.*

This is already present in Figure 3D

| (8) *Testing whether receptor shedding or conformational masking contributes to immune evasion.*

A lifetime’s work

| (9) *Evolutionary trade-offs: Discuss why *T. brucei* maintains ~15 TfR variants if the GPI-anchor number has minimal impact on function (Figure 3).*

The possible reason for the evolution of ~15 TfR variants was discussed in a previous publication.

| (10) *How do their findings align with recent studies on ISG75 surface exposure?*

If this refers to the finding that ISG75 is an Ig Fc receptor, this has been included

| (11) *Add scale bars to 3D reconstructions (Figure 5).*

Added

| (12) *Include a schematic summarizing key findings in the main text.*

Chosen not to do

| (13) *Explicitly state where raw microscopy images, flow cytometry data, and analysis scripts are deposited.*

Microscope Images have deposited in Bioimage Archive repository at EMBL/EBI No flow cytometry used

| (14) *Correct inconsistent GPI-anchor terminology (e.g., "glycosylphosphoinositol" to "glycosylphosphatidylinositol").*

Our typo, corrected

| (15) Clarify ambiguous phrases (e.g., "subtle mechanisms" in the Discussion).

Corrected

<https://doi.org/10.7554/eLife.107191.2.sa0>